INCH-POUND

MIL-M-28787/368(SH)

31 July 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

DYNAMIC RAM CONTROLLER MODULE, KEY CODE FER

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic dynamic Random Access Memory (RAM) module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.1662 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be FER.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 130 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	S	1	INF	1	INF	1	INF	MOhm
A2-A33	Isolation, initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	X	1	INF	1	INF	1	INF	MOhm
B1-B16	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B17-B32	Continuity	III	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Vcc stress	IV	Q	Module shall pass all remaining tests						
D1-D15	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D16-D24	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D25	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D26,D27	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
E1-E4	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V
E5-E27	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
F1-F4	Ios	VII	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
F5-F17	Ios	VII	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
F18	Ios	VII	S	-250.25	-30.0	-250.25	-30.0	-250.25	-30.0	mA
F19	Ios	VII	S	-100.25	-40.0	-100.25	-40.0	-100.25	-40.0	mA
F20	Ios	VII	S	-130.4	-15.03	-130.4	-15.03	-130.4	-15.03	mA
F21	Ios	VII	S	-132.0	-18.0	-132.0	-18.0	-132.0	-18.0	mA
F22	Ios	VII	S	-253.76	-31.0	-253.76	-31.0	-253.76	-31.0	mA
G1-G4	Iih	VIII	X	0	50	0	50	0	50	uA
G5-G15	Iih	VIII	X	0	25	0	25	0	25	uA
G16-G24	Iih	VIII	X	0	20	0	20	0	20	uA
G25	Iih	VIII	X	0	80	0	80	0	80	uA
G26,G27	Iih	VIII	X	0	50	0	50	0	50	uA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-2/	25 ± 5°C 3/		0 ± 5°C 4/		55 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
H1-H4	Iil	IX	X	-2.0	-1.0	-2.0	-1.0	-2.0	-1.0	mA
H5-H15	Iil	IX	X	-0.25	0.0	-0.25	0.0	-0.25	0.0	mA
H16-H24	Iil	IX	X	-0.4	-0.03	-0.4	-0.03	-0.4	-0.03	mA
H25	Iil	IX	X	-1.5	-0.06	-1.5	-0.06	-1.5	-0.06	mA
H26,H27	Iil	IX	X	-0.5	0.0	-0.5	0.0	-0.5	0.0	mA
J1-J4	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
J5-J18	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J19-J21	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
J22	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J23	Voh	X	X	2.4	2.6	2.4	2.6	2.4	2.6	V
K1-K19	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K20,K21	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K22	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L	Icc	XII	X	50	570	50	570	50	570	mA
M1,M2	Functional	XIII	X	See table XIII						
N1,N2	Dynamic	XIV	S	See table XIV						
P	PSFC	XVI	S	3.9	6.5	3.9	6.5	3.9	6.5	uF
Q	PSTR	XVII	Q	0	400	0	400	0	400	mV
R1-R18	Output crosstalk (high level)	XVIII	Q	0	500	0	500	0	500	mV
S1-S4	Output crosstalk (low level)	XIX	Q	0	750	0	750	0	750	mV
S5-S18	Output crosstalk (low level)	XIX	Q	0	500	0	500	0	500	mV

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
T1-T8	Cin	XX	Q	2	25	2	25	2	25	pF
T9-T12	Cin	XX	Q	2	30	2	30	2	30	pF
T13-T19	Cin	XX	Q	2	20	2	20	2	20	pF
T20,T21	Cin	XX	Q	2	31	2	31	2	31	pF
T22,T23	Cin	XX	Q	4	80	4	80	4	80	pF
T24-T26	Cin	XX	Q	2	35	2	35	2	35	pF
T27	Cin	XX	Q	4	60	4	80	4	60	pF
U1	Cout	XXI	Q	4	140	4	140	4	140	pF
U2-U4	Cout	XXI	Q	2	45	2	45	2	45	pF
U5-U8	Cout	XXI	Q	2	30	2	30	2	30	pF
U8-U15	Cout	XXI	Q	2	20	2	20	2	20	pF
U16	Cout	XXI	Q	2	38	2	38	2	38	pF
U17,U18	Cout	XXI	Q	2	95	2	95	2	95	pF
U19	Cout	XXI	Q	2	25	2	25	2	25	pF
U20	Cout	XXI	Q	2	30	2	30	2	30	pF
U21	Cout	XXI	Q	2	60	2	60	2	60	pF
U22	Cout	XXI	Q	2	70	2	70	2	70	pF
V	Threshold	XXII	X	See table XXII						
W1-W4	Output crosstalk duration	XXIII	Q	0	59	0	63	0	62	nS

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/368-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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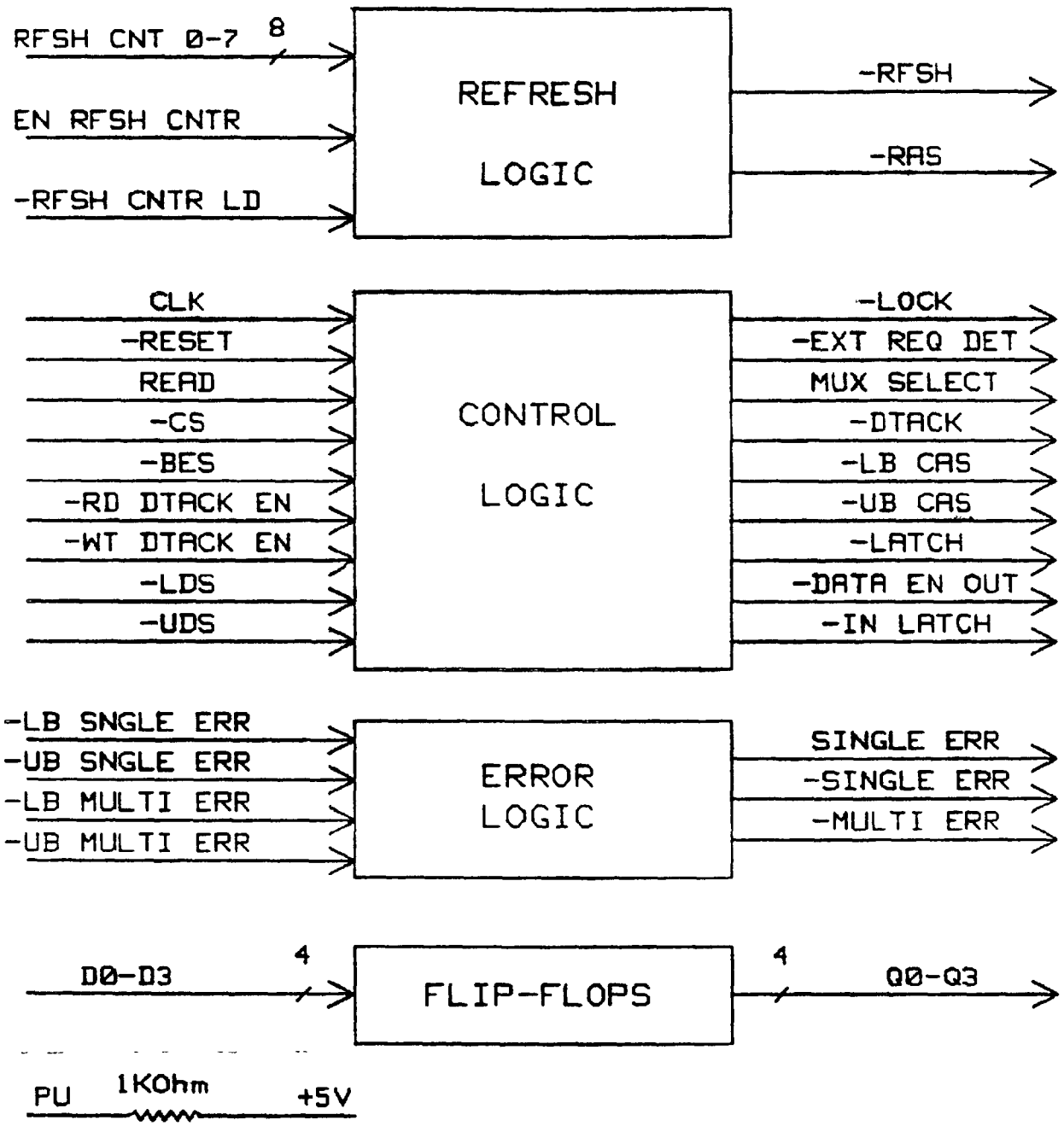


FIGURE 1. FER functional block diagram.

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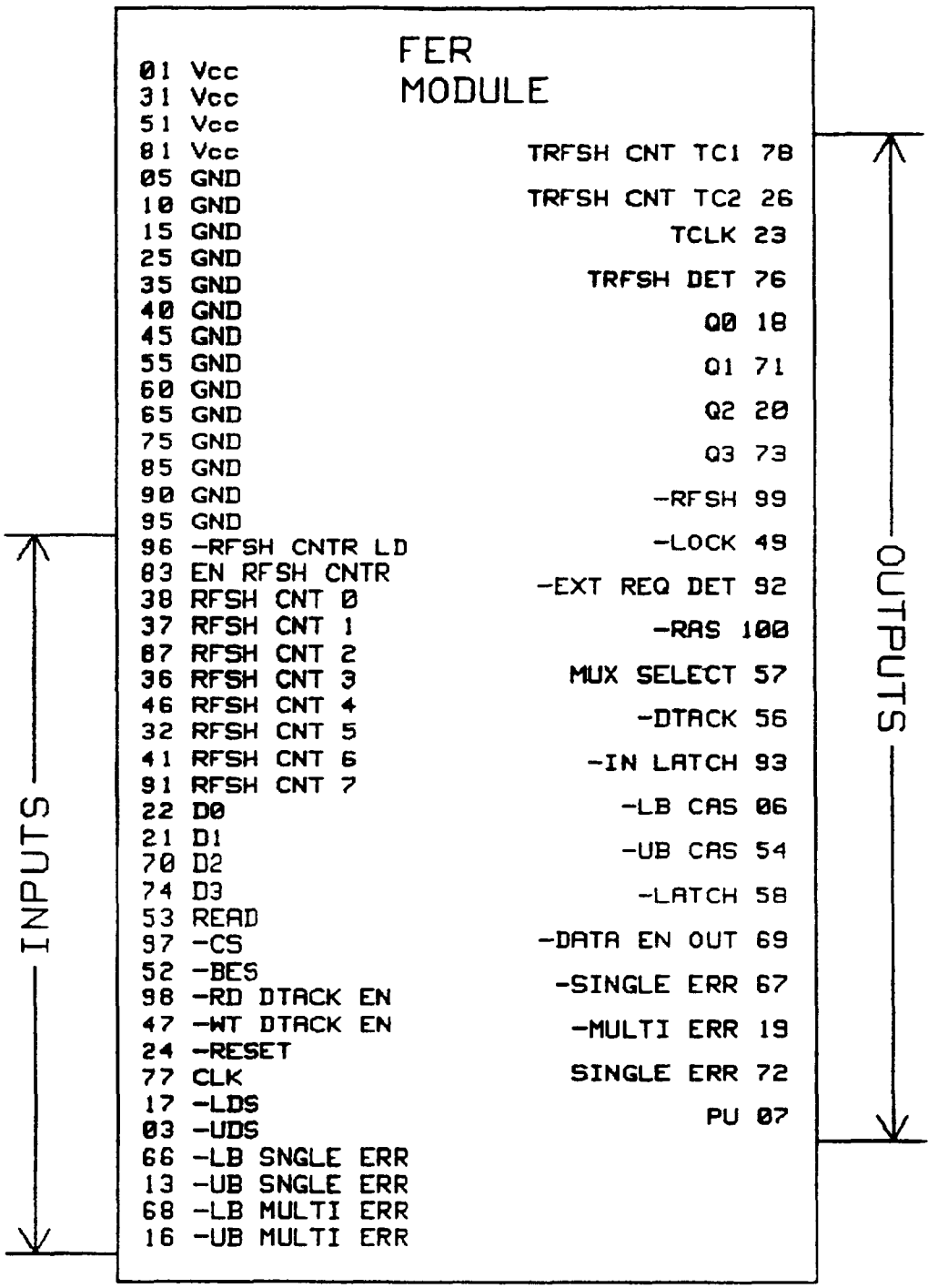


FIGURE 2. FER pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

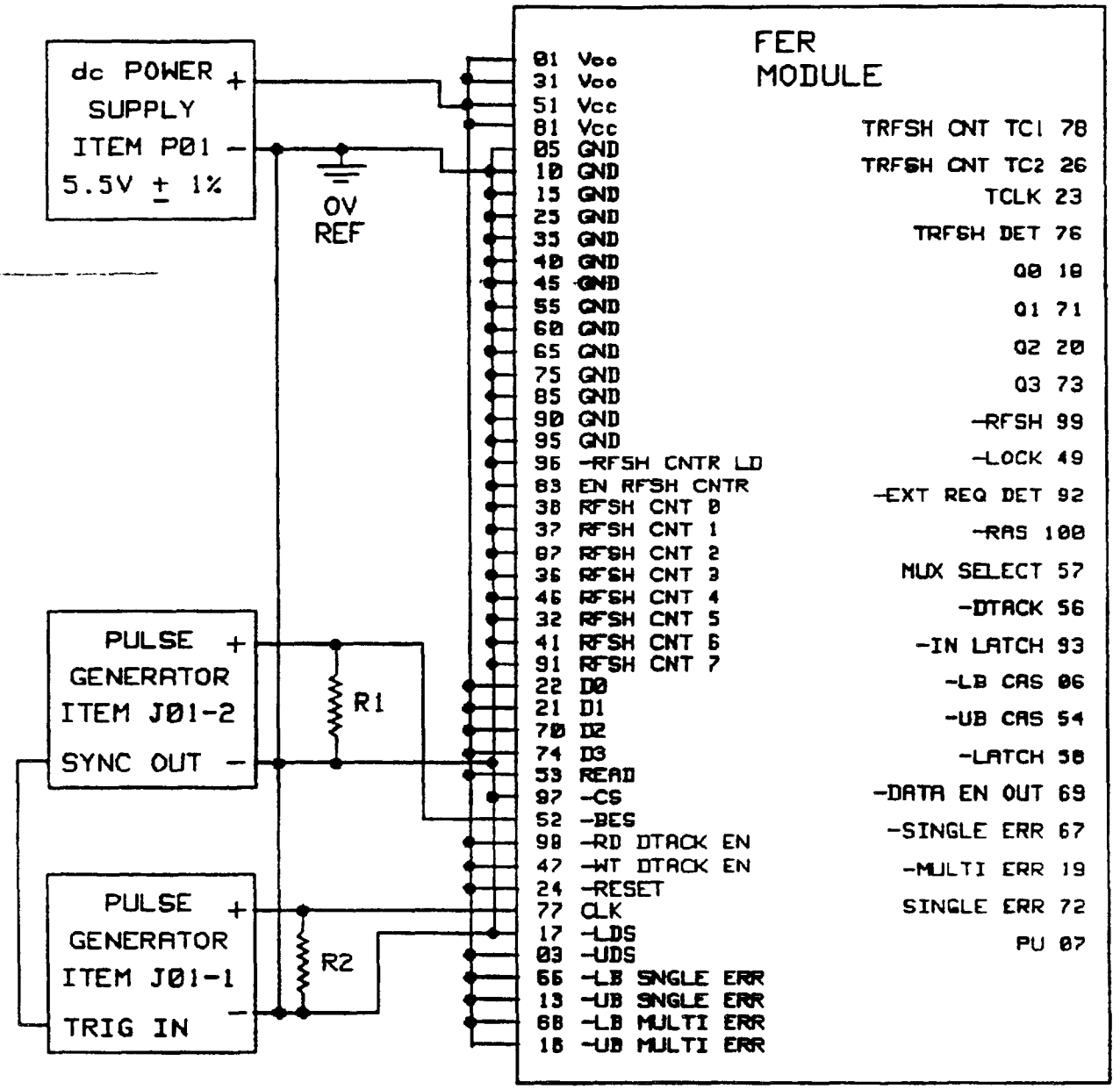
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through S, V and W shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figures 3 and 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 5 and 6.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 7. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition P, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. R1 = R2 = 51 ohm ± 5 percent, 1W.

FIGURE 3. FER operating temperature and life test circuit.

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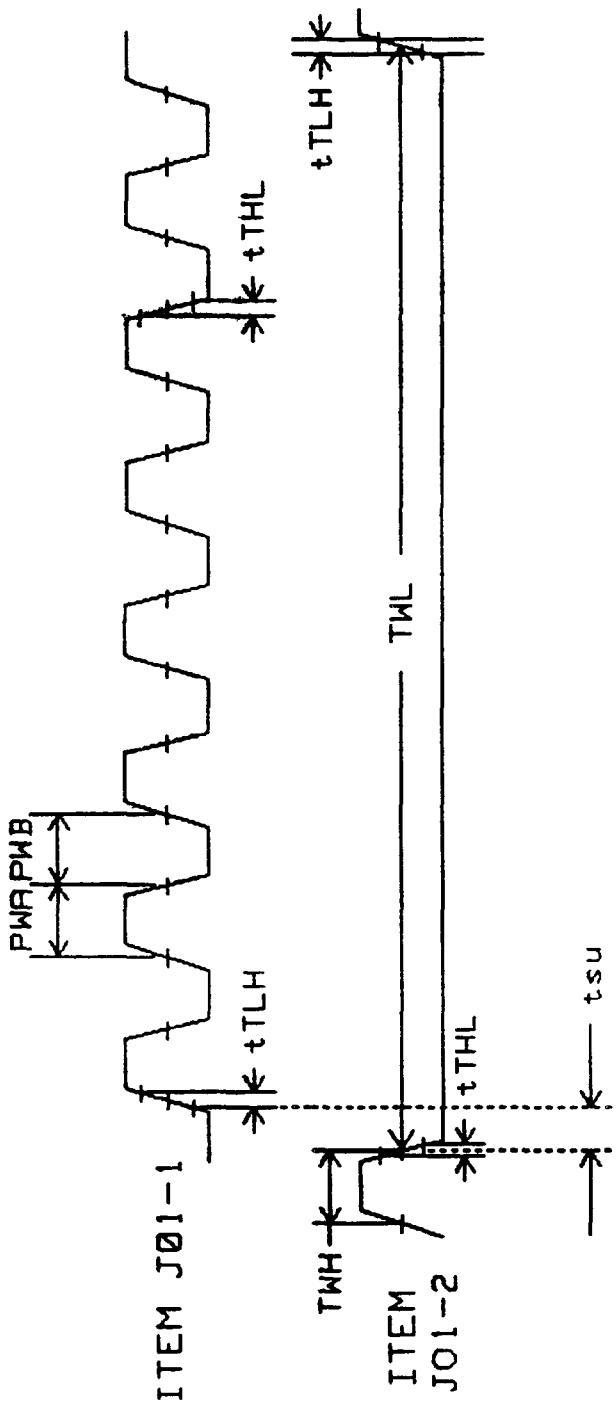


FIGURE 4. Operating temperature and life test waveforms.

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NOTES:

1. Input pulse parameters for item J01-1:

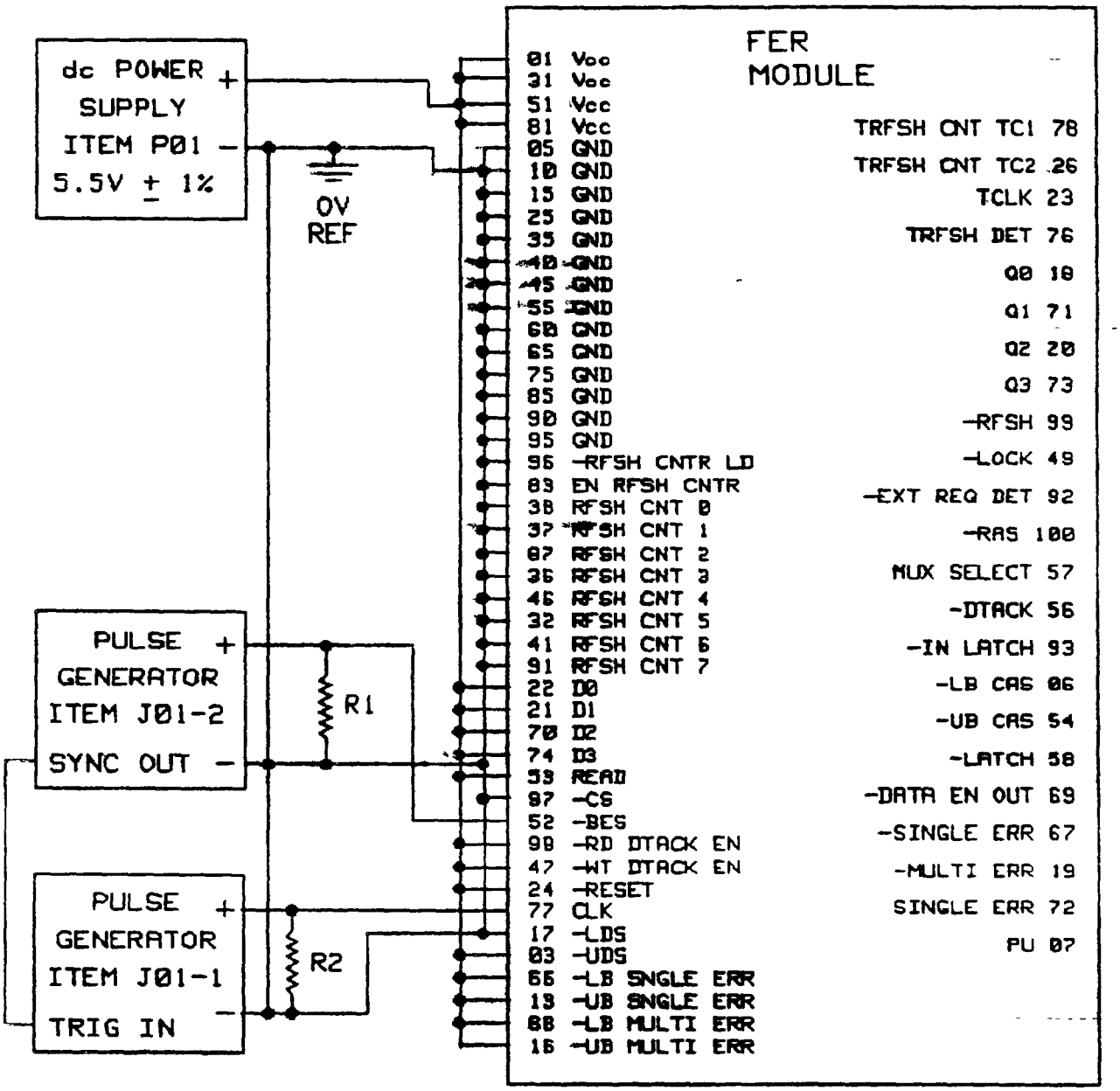
   PWA = 250 nS minimum.
   PWB = 250 nS minimum.
   tTHL = tTLH = 50 ± 10 nS.

2. Input pulse parameters for item J01-2:

   tTHL = tTLH = 50 ± 10 nS.
   TWL = ((PWA + PWB) x 8 + 30) nS.
   TWH = 50 nS minimum.

3. tsu = 50 ± 10 nS.

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NOTE:

1. R1 = R2 = 51 ohm ± 5 percent, 1W.

FIGURE 5. FER component temperature test circuit.

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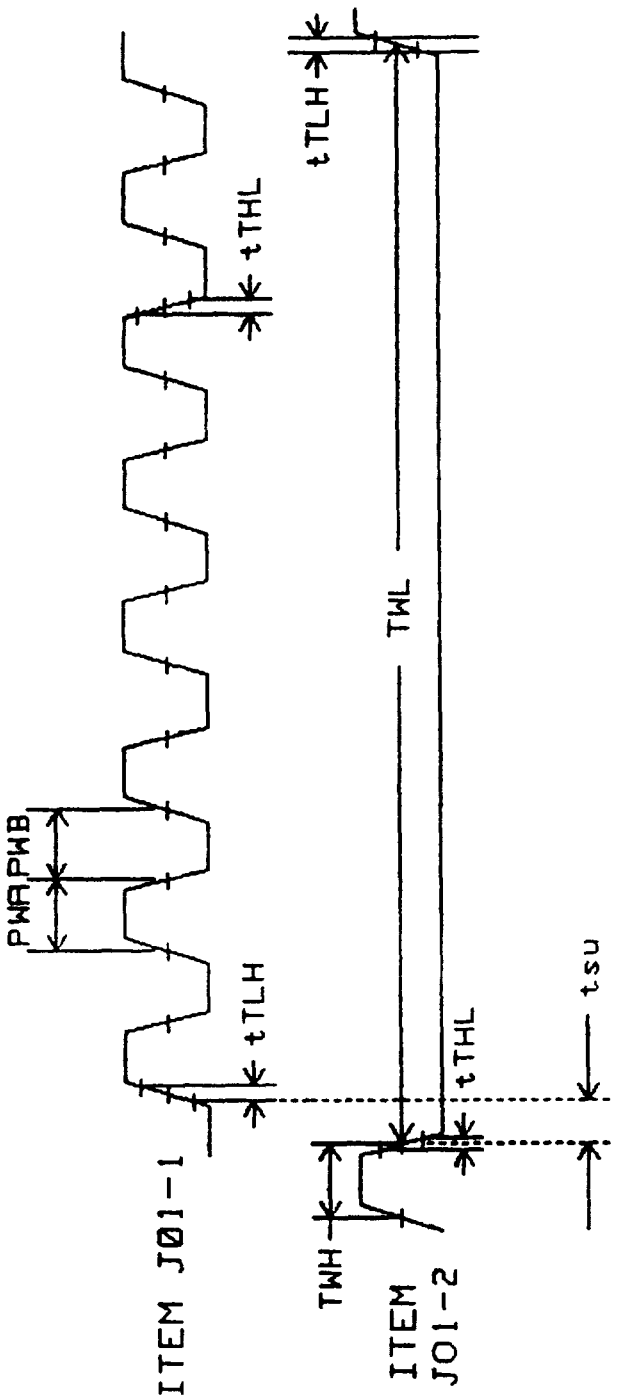


FIGURE 6. Component temperature test waveforms.

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NOTES:

1. Input pulse parameters for item J01-1:

   PWA = 250 nS minimum.
   PWB = 250 nS minimum.
   tTHL = tTLH = 50 ± 10 nS.

2. Input pulse parameters for item J01-2:

   tTHL = tTLH = 50 ± 10 nS.
   TWL = ((PWA + PWB) x 8 + 30) nS.
   TWH = 50 nS minimum.

3. tsu = 50 ± 10 nS.

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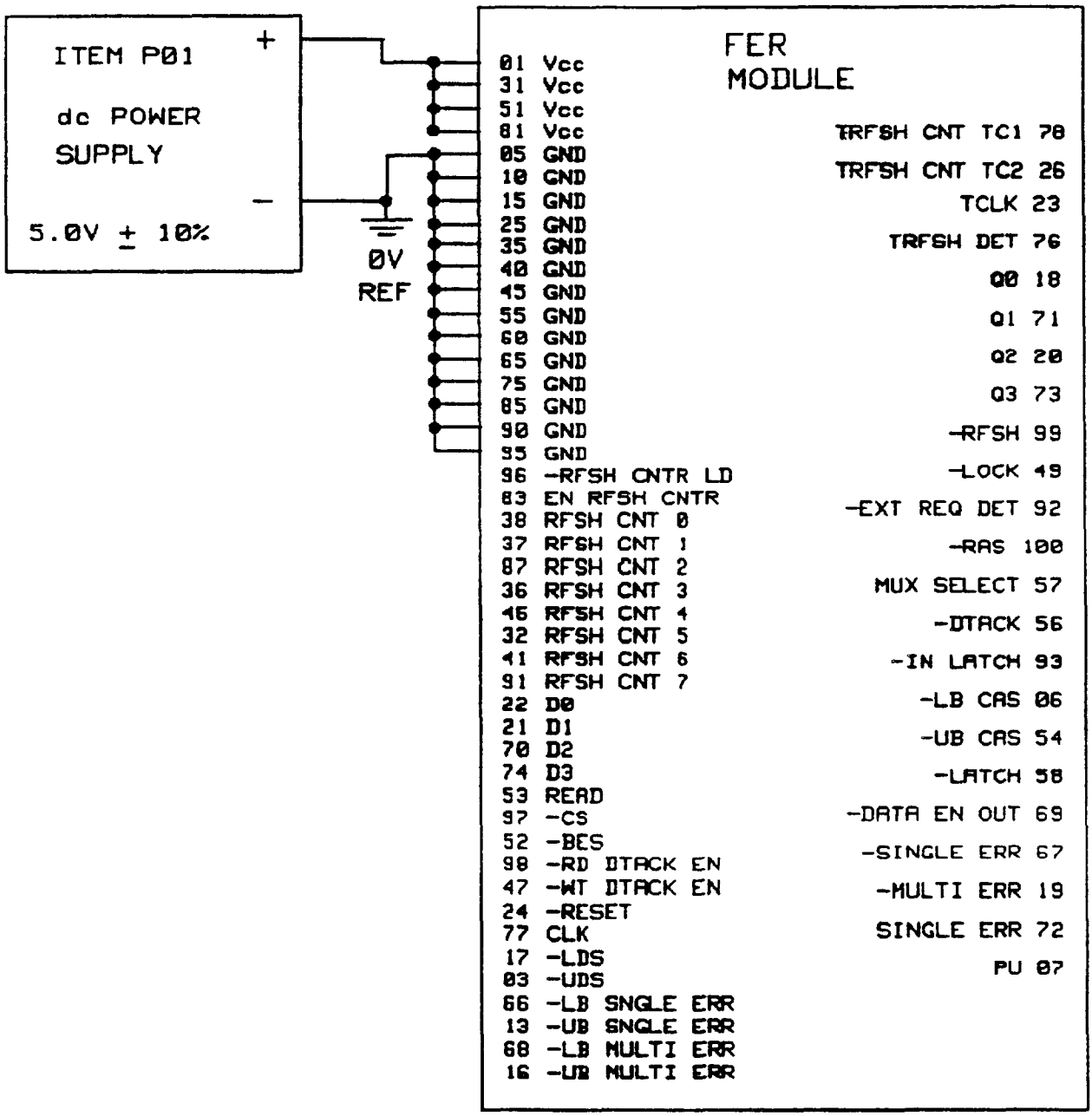


FIGURE 7. FER vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests B, G through S, V and W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through S, V and W shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) NA = not applicable.
(l) PSFC = power supply filter capacitance.
(m) PSTA = power supply transient amplitude.
(n) Vic = input clamp diode voltage.
(o) Vih = input high level voltage.
(p) Vil = input low level voltage.
(q) Vin = input forcing voltage.
(r) Voh = output high level voltage.
(s) Vol = output low level voltage.
(t) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 3.185 W (max); 1.999 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
RFSH CNT 0	-1.5	0.7	2.0	-0.4	20	100	25
RFSH CNT 1	-1.5	0.7	2.0	-0.4	20	100	25
RFSH CNT 2	-1.5	0.7	2.0	-0.4	20	100	25
RFSH CNT 3	-1.5	0.7	2.0	-0.4	20	100	25
EN RFSH CNTR	-1.5	0.7	2.0	-0.4	20	100	25
D0	-1.2	0.8	2.0	-2.0	50	1000	20
D1	-1.2	0.8	2.0	-2.0	50	1000	20
D2	-1.2	0.8	2.0	-2.0	50	1000	20
D3	-1.2	0.8	2.0	-2.0	50	1000	20
RFSH CNT 4	-1.5	0.7	2.0	-0.4	20	100	30
RFSH CNT 5	-1.5	0.7	2.0	-0.4	20	100	30
RFSH CNT 6	-1.5	0.7	2.0	-0.4	20	100	30
RFSH CNT 7	-1.5	0.7	2.0	-0.4	20	100	30
-RFSH CNTR LD	-1.5	0.7	2.0	-1.6	80	400	60

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
READ	-1.5	0.8	2.0	-0.5	50	2000	80
-BES	-1.5	0.8	2.0	-0.5	50	2000	80
-CS	-1.5	0.8	2.0	-0.25	25	1000	25
-WT DTACK EN	-1.5	0.8	2.0	-0.25	25	1000	25
-UDS	-1.5	0.8	2.0	-0.25	25	1000	25
-LDS	-1.5	0.8	2.0	-0.25	25	1000	20
-LB SNGLE ERR	-1.5	0.8	2.0	-0.25	25	1000	20
-UB SNGLE ERR	-1.5	0.8	2.0	-0.25	25	1000	20
CLK	-1.5	0.8	2.0	-0.25	25	1000	31
-LB MULTI ERR	-1.5	0.8	2.0	-0.25	25	1000	31
-UB MULTI ERR	-1.5	0.8	2.0	-0.25	25	1000	35
-RESET	-1.5	0.8	2.0	-0.25	25	1000	35
-RD DTACK EN	-1.5	0.8	2.0	-0.25	25	1000	35

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
Q0	2.5	0.5	-1000	20.0	-100.0	20
Q1	2.5	0.5	-1000	20.0	-100.0	20
Q2	2.5	0.5	-1000	20.0	-100.0	20
Q3	2.5	0.5	-1000	20.0	-100.0	20
TRFSH DET	2.5	0.5	-975	19.75	-100.25	45
TRFSH CNT TC1	2.5	0.4	-380	3.6	-130.4	45
TRFSH CNT TC2	2.5	0.4	-350	2.0	-132.0	38
TCLK	2.4	0.5	-1820	8.24	-253.76	140
-RFSH	2.4	0.5	-2000	12.0	-250.0	20
-LOCK	2.4	0.5	-2000	12.0	-250.0	20
-RAS	2.4	0.5	-2000	12.0	-250.0	20
-MULTI ERR	2.4	0.5	-2000	12.0	-250.0	30
-DATA EN OUT	2.4	0.5	-2000	12.0	-250.0	30

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (uA)	Iol max (mA)	Ios max (mA)	Cout max (pF)
-LATCH	2.4	0.5	-2000	12.0	-250.0	30
-LB CAS	2.4	0.5	-2000	12.0	-250.0	30
-IN LATCH	2.4	0.5	-2000	12.0	-250.0	35
-UB CAS	2.4	0.5	-2000	12.0	-250.0	35
-EXT REQ DET	2.4	0.5	-2000	12.0	-250.0	30
-DTACK	2.4	0.5	-2000	12.0	-250.0	60
-SINGLE ERR	2.4	0.5	-2000	12.0	-250.0	25
SINGLE ERR	2.4	0.5	-2000	12.0	-250.0	45
MUX SELECT	2.4	0.5	-1975	11.75	-250.25	70
PU	2.4	NA	-2200	NA	NA	NA

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The module function is not printed herein because of its length. If required, a copy may be obtained from Naval Weapons Support Center (Code 602).

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6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
D0 D1 D2 D3	22 21 70 74	I I I I	These are general purpose Schottky D-type flip-flop data inputs.
-CS	97	I	The active low chip select input is used with -BES to condition the start of a Dynamic RAM (DRAM) read or write cycle.
-RD DTACK EN	98	I	When low during a read cycle, this input conditions the assertion of the Data Transfer Acknowledge output (-DTACK).
-WT DTACK EN	47	I	When low during a write cycle, this input conditions the assertion of -DTACK.
-RESET	24	I	This active low input is the system reset signal.
CLK	77	I	This provides the clock for all devices on the module.
-LDS	17	I	This active low signal is used to condition the assertion of the Lower Byte Column Address Strobe (-LB CAS).
-UDS	3	I	This active low signal is used to condition the assertion of the Upper Byte Column Address Strobe (-UB CAS).
-LB SNGLE ERR	66	I	This input signal, when active low, indicates that a single error has been detected in the current DRAM lower byte data.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-UB SNGLE ERR	13	I	This input signal, when active low, indicates that a single error has been detected in the current DRAM upper byte data.
-LB MULTI ERR	68	I	This input signal, when active low, indicates that a multiple error has been detected in the current DRAM lower byte data.
-UB MULTI ERR	16	I	This input signal, when active low, indicates that a multiple error has been detected in the current DRAM upper byte data.
RFSH CNT 0 RFSH CNT 1 RFSH CNT 2 RFSH CNT 3 RFSH CNT 4 RFSH CNT 5 RFSH CNT 6 RFSH CNT 7	38 37 87 36 46 32 41 91	I I I I I I I I	These inputs are used to determine the frequency of refresh for the DRAMs. These inputs are either tied to ground or pulled up to Vcc depending on the type of DRAM used. They provide the module with a programmable refresh frequency.
EN RFSH CNTR	83	I	When active (high), the refresh count is enabled.
-RFSH CNTR LD	96	I	A low level at this input disables the refresh count and causes the data at the RFSH CNT inputs to be loaded on the next positive going edge of the clock.
READ	53	I	When this input is high, the DRAM is to be read by the microprocessor. A low indicates a write operation.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-BES	52	I	This signal, when active (low), is used with -CS to condition the start of a DRAM read or write cycle.
Q0 Q1 Q2 Q3	18 71 20 73	O O O O	These are general purpose Schottky D-type flip-flop outputs.
-RFSH	99	O	This signal, when active (low), indicates that a DRAM refresh cycle is occurring.
-LOCK	49	O	When active (low), this signal shows that a memory cycle request has been serviced. It stretches out the cycle to allow a refresh during a memory cycle.
-EXT REQ DET	92	O	This signal becomes active (low) when -BES and -CS are low, thus detecting a DRAM memory cycle request.
-RAS	100	O	This active low signal provides a Row Address Strobe for the DRAM memory array.
-DTACK	56	O	This active low output signal indicates to the microprocessor that the DRAM bus transfer cycle can be completed.
-IN LATCH	93	O	This active low signal is used to latch DRAM array read data into the error detection and correction unit in the DFH module.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-LB CAS	6	O	This active low output activates the Lower Byte DRAM Column Address Strobe unless a refresh cycle is in progress.
-UB CAS	54	O	This active low output activates the Upper Byte DRAM Column Address Strobe unless a refresh cycle is in progress.
-LATCH	58	O	This signal provides internal FER timing sequence control.
-DATA EN OUT	69	O	This active low signal is sent to the error detection and correction unit in the DFH module to enable the data output latch.
-SINGLE ERR	67	O	This active low output signal indicates that a single error exists in the current DRAM contents.
SINGLE ERR	72	O	This active high signal is the complement of -SINGLE ERR (false output).
-MULTI ERR	19	O	This active low output signal indicates that a multiple error exists in the current DRAM data.
MUX SELECT	57	O	This active high output determines whether row or column address inputs appear at the multiplexer outputs of the address controller in the DFH module.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
TRFSH DET	76	O	Test point: This active high signal initiates a refresh cycle for the DRAM. It is observed as an output test point only.
TRFSH CNT TC1	78	O	Test point: Refresh counter 1 terminal count. This signal goes high when the first refresh counter reaches a count of all 1's. This enables the second refresh counter to count.
TRFSH CNT TC2	26	O	Test point: Refresh counter 2 terminal count. This signal goes high when the second refresh counter reaches a count of all 1's. This allows a refresh cycle to occur after 3 more clock delays.
TCLK	23	O	Test point: This signal provides a clock input to all devices on the module. It is observed as an output test point only.
PU	7	O	This output is a connection to an internal 1K ohm pull-up resistor. It is available for general purpose use in tying module inputs to a logic high.

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 8.

(j) Vol test - table XI, figure 8.

(k) Icc test - table XII.

(l) Functional test - table XIII, figure 9.

(m) Dynamic test - tables XIV and XV, figures 10 and 11.

(n) PSFC test - table XVI.

(o) PSTA test - table XVII, figure 12.

(p) Output crosstalk tests - tables XVIII and XIX, figures 13 and 14.

(q) Cin test - table XX.

(r) Cout test - table XXI.

(s) Threshold test - table XXII.

(t) Output crosstalk duration test - table XXIII, figure 15.

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6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N747)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	All pins	Frame	10	INF	MOhm	1-7
A2	Isolation, used to unused pins	All used pins	All unused pins	10	INF	MOhm	1-7
A3	Isolation, unused to unused pins	94	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59,61,62,63,64,79, 80,82,84,86,88,89	10	INF	MOhm	1-7
A4	Isolation, unused to unused pins	89	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59,61,62,63,64,79, 80,82,84,86,88	10	INF	MOhm	1-7
A5	Isolation, unused to unused pins	88	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59,61,62,63,64,79, 80,82,84,86	10	INF	MOhm	1-7
A6	Isolation, unused to unused pins	86	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59,61,62,63,64,79, 80,82,84	10	INF	MOhm	1-7
A7	Isolation, unused to unused pins	84	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59,61,62,63,64,79, 80,82	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A8	Isolation, unused to unused pins	82	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59,61,62,63,64,79, 80	10	INF	MOhm	1-7
A9	Isolation, unused to unused pins	80	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59,61,62,63,64,79	10	INF	MOhm	1-7
A10	Isolation, unused to unused pins	79	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59,61,62,63,64	10	INF	MOhm	1-7
A11	Isolation, unused to unused pins	64	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59,61,62,63	10	INF	MOhm	1-7
A12	Isolation, unused to unused pins	63	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59,61,62	10	INF	MOhm	1-7
A13	Isolation, unused to unused pins	62	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59,61	10	INF	MOhm	1-7
A14	Isolation, unused to unused pins	61	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50, 59	10	INF	MOhm	1-7
A15	Isolation, unused to unused pins	59	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48,50	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A16	Isolation, unused to unused pins	50	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44,48	10	INF	MOhm	1-7
A17	Isolation, unused to unused pins	48	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43,44	10	INF	MOhm	1-7
A18	Isolation, unused to unused pins	44	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42,43	10	INF	MOhm	1-7
A19	Isolation, unused to unused pins	43	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39,42	10	INF	MOhm	1-7
A20	Isolation, unused to unused pins	42	2,4,8,9,11,12,14, 27,28,29,30,33,34, 39	10	INF	MOhm	1-7
A21	Isolation, unused to unused pins	39	2,4,8,9,11,12,14, 27,28,29,30,33,34	10	INF	MOhm	1-7
A22	Isolation, unused to unused pins	34	2,4,8,9,11,12,14, 27,28,29,30,33	10	INF	MOhm	1-7
A23	Isolation, unused to unused pins	33	2,4,8,9,11,12,14, 27,28,29,30	10	INF	MOhm	1-7
A24	Isolation, unused to unused pins	30	2,4,8,9,11,12,14, 27,28,29	10	INF	MOhm	1-7
A25	Isolation, unused to unused pins	29	2,4,8,9,11,12,14, 27,28	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A26	Isolation, unused to unused pins	28	2,4,8,9,11,12,14, 27	10	INF	MOhm	1-7
A27	Isolation, unused to unused pins	27	2,4,8,9,11,12,14	10	INF	MOhm	1-7
A28	Isolation, unused to unused pins	14	2,4,8,9,11,12	10	INF	MOhm	1-7
A29	Isolation, unused to unused pins	12	2,4,8,9,11	10	INF	MOhm	1-7
A30	Isolation, unused to unused pins	11	2,4,8,9	10	INF	MOhm	1-7
A31	Isolation, unused to unused pins	9	2,4,8	10	INF	MOhm	1-7
A32	Isolation, unused to unused pins	8	2,4	10	INF	MOhm	1-7
A33	Isolation, unused to unused pins	4	2	10	INF	MOhm	1-7

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V 30 V	± 10 percent ± 10 percent	A1 A2-A33
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A1-A33 A1-A33

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, Vcc	1	31	0.0	1.0	Ohm	1-4
B2	Continuity, Vcc	31	51				
B3	Continuity, Vcc	51	81				
B4	Continuity, GND	5	10				
B5	Continuity, GND	10	15				
B6	Continuity, GND	15	25				
B7	Continuity, GND	25	35				
B8	Continuity, GND	35	40				
B9	Continuity, GND	40	45				
B10	Continuity, GND	45	55				
B11	Continuity, GND	55	60				
B12	Continuity, GND	60	65				
B13	Continuity, GND	65	75				
B14	Continuity, GND	75	85				
B15	Continuity, GND	85	90				
B16	Continuity, GND	90	95	0.0	1.0	Ohm	1-4
B17	Continuity, Vcc	1	31	0.0	0.1	Ohm	1-4
B18	Continuity, Vcc	31	51				
B19	Continuity, Vcc	51	81				
B20	Continuity, GND	5	10	0.0	0.1	Ohm	1-4

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TABLE III. Continuity test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B21	Continuity, GND	10	15	0.0	0.1	Ohm	1-4
B22	Continuity, GND	15	25				
B23	Continuity, GND	25	35				
B24	Continuity, GND	35	40				
B25	Continuity, GND	40	45				
B26	Continuity, GND	45	55				
B27	Continuity, GND	55	60				
B28	Continuity, GND	60	65				
B29	Continuity, GND	65	75				
B30	Continuity, GND	75	85				
B31	Continuity, GND	85	90				
B32	Continuity, GND	90	95	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B16 B17-B32

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NOTES: - Continued.

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31, 51,81	5,10, 15,25, 35,40, 45,55, 60,65, 75,85, 90,95	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	0.896 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
D1	D0	22	5.5	0.0	1.0	mA	1-3
D2	D1	21					
D3	D2	70					
D4	D3	74					
D5	-CS	97					
D6	-RD DTACK EN	98					
D7	-WT DTACK EN	47					
D8	-RESET	24					
D9	CLK	77					
D10	-LDS	17					
D11	-UDS	3					
D12	-LB SNGLE ERR	66					
D13	-UB SNGLE ERR	13					
D14	-LB MULTI ERR	68					
D15	-UB MULTI ERR	16	5.5	0.0	1.0	mA	1-3
D16	RFSH CNT 0	38	5.5	0.0	0.1	mA	1-3
D17	RFSH CNT 1	37					
D18	RFSH CNT 2	87					
D19	RFSH CNT 3	36					
D20	RFSH CNT 4	46	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	RFSH CNT 5	32	5.5	0.0	0.1	mA	1-3
D22	RFSH CNT 6	41					
D23	RFSH CNT 7	91					
D24	EN RFSH CNTR	83	5.5	0.0	0.1	mA	1-3
D25	-RFSH CNTR LD	96	5.5	0.0	0.4	mA	1-3
D26	READ	53	5.5	0.0	2.0	mA	1-3
D27	-BES	52	5.5	0.0	2.0	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D27
Power supply current limit	0.868 A	NA	D1-D27
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D27
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D27
low, (VL)	0.0 V	± 100 mV	D1-D27

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 50 uA	D1-D15
	NA	± 5 uA	D16-D24
	NA	± 20 uA	D25
	NA	± 100 uA	D26,D27

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	D0	22	-18.0	-1.2	-0.4	V	1-3
E2	D1	21					
E3	D2	70					
E4	D3	74	-18.0	-1.2	-0.4	V	1-3
E5	-CS	97	-18.0	-1.5	-0.4	V	1-3
E6	-RD DTACK EN	98					
E7	-WT DTACK EN	47					
E8	-RESET	24					
E9	CLK	77					
E10	-LDS	17					
E11	-UDS	3					
E12	-LB SNGLE ERR	66					
E13	-UB SNGLE ERR	13					
E14	-LB MULTI ERR	68					
E15	-UB MULTI ERR	16					
E16	RFSH CNT 0	38					
E17	RFSH CNT 1	37					
E18	RFSH CNT 2	87					
E19	RFSH CNT 3	36					
E20	RFSH CNT 4	46	-18.0	-1.5	-0.4	V	1-3

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	RFSH CNT 5	32	-18.0	-1.5	-0.4	V	1-3
E22	RFSH CNT 6	41					
E23	RFSH CNT 7	91					
E24	EN RFSH CNTR	83					
E25	-RFSH CNTR LD	96					
E26	READ	53					
E27	-BES	52	-18.0	-1.5	-0.4	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E27
Power supply current limit	0.868 A	NA	E1-E27
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E27
Pattern voltage high, (VH)	3.0 V	± 100 mV	E1-E27
low, (VL)	0.0 V	± 100 mV	E1-E27

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage measurement	NA NA	± 40 mV ± 55 mV	E1-E4 E5-E27

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	Q0	18	0.0	-100.0	-40.0	mA	1-4
F2	Q1	71					
F3	Q2	20					
F4	Q3	73	0.0	-100.0	-40.0	mA	1-4
F5	-RFSH	99	0.0	-250.0	-30.0	mA	1-4
F6	-LOCK	49					
F7	-EXT REQ DET	92					
F8	-RAS	100					
F9	-DTACK	56					
F10	-IN LATCH	93					
F11	-LB CAS	6					
F12	-UB CAS	54					
F13	-LATCH	58					
F14	-DATA EN OUT	69					
F15	-SINGLE ERR	67					
F16	SINGLE ERR	72					
F17	-MULTI ERR	19	0.0	-250.0	-30.0	mA	1-4
F18	MUX SELECT	57	0.0	-250.25	-30.0	mA	1-4
F19	TRFSH DET	76	0.0	-100.25	-40.0	mA	1-4
F20	TRFSH CNT TC1	78	0.0	-130.4	-15.03	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	TRFSH CNT TC2	26	0.0	-132.0	-16.0	mA	1-4
F22	TCLK	23	0.0	-253.76	-31.0	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F22
Power supply current limit	0.868 A	NA	F1-F22
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F22
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F22 F1-F22
Output current measurement	NA NA NA NA NA NA NA	± 3.0 mA ± 11.0 mA ± 11.0 mA ± 3.01 mA ± 5.77 mA ± 5.8 mA ± 11.1 mA	F1-F4 F5-F17 F18 F19 F20 F21 F22

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	D0	22	2.7	0	50	uA	1-3
G2	D1	21					
G3	D2	70					
G4	D3	74	2.7	0	50	uA	1-3
G5	-CS	97	2.4	0	25	uA	1-3
G6	-RD DTACK EN	98					
G7	-WT DTACK EN	47					
G8	-RESET	24					
G9	CLK	77					
G10	-LDS	17					
G11	-UDS	3					
G12	-LB SNGLE ERR	66					
G13	-UB SNGLE ERR	13					
G14	-LB MULTI ERR	68					
G15	-UB MULTI ERR	16	2.4	0	25	uA	1-3
G16	RFSH CNT 0	38	2.7	0	20	uA	1-3
G17	RFSH CNT 1	37					
G18	RFSH CNT 2	87					
G19	RFSH CNT 3	36					
G20	RFSH CNT 4	46	2.7	0	20	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	RFSH CNT 5	32	2.7	0	20	uA	1-3
G22	RFSH CNT 6	41					
G23	RFSH CNT 7	91					
G24	EN RFSH CNTR	83	2.7	0	20	uA	1-3
G25	-RFSH CNTR LD	96	2.7	0	80	uA	1-3
G26	READ	53	2.4	0	50	uA	1-3
G27	-BES	52	2.4	0	50	uA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G27
Power supply current limit	0.868 A	NA	G1-G27
Forcing voltage, (Vih)	2.7 V 2.4 V 2.7 V 2.4 V	± 135 mV ± 120 mV ± 135 mV ± 120 mV	G1-G4 G5-G15 G16-G25 G26,G27
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G27 G1-G27

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 2.5 uA	G1-G4
	NA	± 1.25 uA	G5-G15
	NA	± 1.0 uA	G16-G24
	NA	± 4.0 uA	G25
	NA	± 2.5 uA	G26,G27

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Initial limits Max	Units	Notes
H1	D0	22	0.5	-2.0	-1.0	mA	1-3
H2	D1	21					
H3	D2	70					
H4	D3	74	0.5	-2.0	-1.0	mA	1-3
H5	-CS	97	0.4	-0.25	0.0	mA	1-3
H6	-RD DTACK EN	98					
H7	-WT DTACK EN	47					
H8	-RESET	24					
H9	CLK	77					
H10	-LDS	17					
H11	-UDS	3					
H12	-LB SNGLE ERR	66					
H13	-UB SNGLE ERR	13					
H14	-LB MULTI ERR	68					
H15	-UB MULTI ERR	16	0.4	-0.25	0.0	mA	1-3
H16	RFSH CNT 0	38	0.4	-0.4	-0.03	mA	1-3
H17	RFSH CNT 1	37					
H18	RFSH CNT 2	87					
H19	RFSH CNT 3	36					
H20	RFSH CNT 4	46	0.4	-0.4	-0.03	mA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	RFSH CNT 5	32	0.4	-0.4	-0.03	mA	1-3
H22	RFSH CNT 6	41					
H23	RFSH CNT 7	91					
H24	EN RFSH CNTR	83	0.4	-0.4	-0.03	mA	1-3
H25	-RFSH CNTR LD	96	0.4	-1.6	-0.06	mA	1-3
H26	READ	53	0.4	-0.5	0.0	mA	1-3
H27	-BES	52	0.4	-0.5	0.0	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H27
Power supply current limit	0.868 A	NA	H1-H27
Forcing voltage, (Vil)	0.5 V 0.4 V	± 25 mV ± 20 mV	H1-H4 H5-H27
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H27 H1-H27

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 50.0 uA	H1-H4
	NA	± 12.5 uA	H5-H15
	NA	± 18.5 uA	H16-H24
	NA	± 77.0 uA	H25
	NA	± 25.0 uA	H26,H27

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	Q0	18	-1.0	2.5	4.5	V	1-5
J2	Q1	71					
J3	Q2	20					
J4	Q3	73	-1.0	2.5	4.5	V	1-5
J5	-RFSH	99	-2.0	2.4	4.5	V	1-5
J6	-LOCK	49					
J7	-EXT REQ DET	92					
J8	-RAS	100					
J9	-DTACK	56					
J10	-IN LATCH	93					
J11	-LB CAS	6					
J12	-UB CAS	54					
J13	-LATCH	58					
J14	-DATA EN OUT	69					
J15	-SINGLE ERR	67					
J16	SINGLE ERR	72					
J17	-MULTI ERR	19	-2.0	2.4	4.5	V	1-5
J18	MUX SELECT	57	-1.975	2.4	4.5	V	1-5
J19	TRFSH DET	76	-0.975	2.5	4.5	V	1-5
J20	TRFSH CNT TC1	78	-0.38	2.5	4.5	V	1-5

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	TRFSH CNT TC2	26	-0.35	2.5	4.5	V	1-5
J22	TCLK	23	-1.82	2.4	4.5	V	1-5
J23	PU	7	-2.0	2.4	2.6	V	1-5

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J23
Power supply current limit	0.868 A	NA	J1-J23
Forcing current, (Ioh)	-1.0 mA	± 10.0 uA	J1-J4
	-2.0 mA	± 20.0 uA	J5-J17
	-1.975 mA	± 19.75 uA	J18
	-0.975 mA	± 9.75 uA	J19
	-0.38 mA	± 3.8 uA	J20
	-0.35 mA	± 3.5 uA	J21
	-1.82 mA	± 18.2 uA	J22
	-2.0 mA	± 20.0 uA	J23
Pattern voltage high, (VH)	See note 4		J1-J23
low, (VL)	See note 4		J1-J23

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage measurement	NA	± 100 mV	J1-J4
	NA	± 105 mV	J5-J18
	NA	± 100 mV	J19-J21
	NA	± 105 mV	J22
	NA	± 10 mV	J23

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

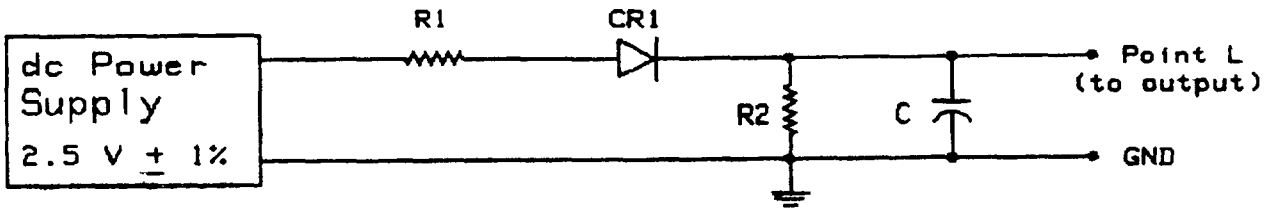
3. The load shall be as indicated on figure 8.

4. The pattern voltages for each input pin are listed below:

Input pin	(VH)	(VL)
38,37,87,36,46,32,41,91,83, 96	2.0 V ± 20 mV	0.7 V ± 7 mV
22,21,70,74,53,97,52,98,47, 24,77,17,3,66,13,68,16	2.0 V ± 20 mV	0.8 V ± 8 mV

5. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 8. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are listed below.

4. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
58,69,56,6,54,49, 99,100,92,93,19, 72,67	1188	12.0	0.5
18,71,20,73	2475	20.0	0.5
57	1203	11.75	0.5
26	7143	2.0	0.4
78	6580	3.6	0.4
23	1320	8.24	0.5
76	2564	19.75	0.5
7	1200	NA	NA

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K1	Q0	18	20.0	0.0	0.5	V	1-5
K2	Q1	71					
K3	Q2	20					
K4	Q3	73	20.0	0.0	0.5	V	1-5
K5	-RFSH	99	12.0	0.0	0.5	V	1-5
K6	-LOCK	49					
K7	-EXT REQ DET	92					
K8	-RAS	100					
K9	-DTACK	56					
K10	-IN LATCH	93					
K11	-LB CAS	6					
K12	-UB CAS	54					
K13	-LATCH	58					
K14	-DATA EN OUT	69					
K15	-SINGLE ERR	67					
K16	SINGLE ERR	72					
K17	-MULTI ERR	19	12.0	0.0	0.5	V	1-5
K18	MUX SELECT	57	11.75	0.0	0.5	V	1-5
K19	TRFSH DET	76	19.75	0.0	0.5	V	1-5
K20	TRFSH CNT TC1	78	3.6	0.0	0.4	V	1-5

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K21	TRFSH CNT TC2	26	2.0	0.0	0.4	V	1-5
K22	TCLK	23	8.25	0.0	0.5	V	1-5

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K22
Power supply current limit	0.868 A	NA	K1-K22
Forcing current, (Iol)	20.0 mA	± 200.0 uA	K1-K4
	12.0 mA	± 120.0 uA	K5-K17
	11.75 mA	± 117.5 uA	K18
	19.75 mA	± 197.5 uA	K19
	3.6 mA	± 36.0 uA	K20
	2.0 mA	± 20.0 uA	K21
	8.25 mA	± 82.5 uA	K22
Pattern voltage high, (VH)	See note 4		K1-K22
low, (VL)	See note 4		K1-K22
Output voltage measurement	NA	± 25 mV	K1-K19
	NA	± 20 mV	K20,K21
	NA	± 25 mV	K22

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 8.

4. The input pattern voltages are listed below:

Input pin	(VH)	(VL)
38,37,87,36,46,32,41,91,83, 96	2.0 V ± 20 mV	0.7 V ± 7 mV
22,21,70,74,53,97,52,98,47, 24,77,17,3,66,13,68,16	2.0 V ± 20 mV	0.8 V ± 8 mV

5. Abbreviations used are defined in 6.1.

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TABLE XII. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
L	Icc	1,31, 51,81	58	579	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L
Power supply current limit	0.868 A	NA	L
Pattern voltage high, (VH)	3.0 V	± 100 mV	L
low, (VL)	0.0 V	± 100 mV	L
Power supply current measurement	NA	± 26 mA	L

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XII.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Functional test.

Test cond	Test description	Notes
M1	Functional test minimum power supply voltage maximum pattern rate	1-3
M2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	M1 M2
Power supply current limit	0.868 A	NA	M1,M2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 M1,M2 M1,M2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 M1 M2 M1,M2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XIII using only a capacitive load of 100 pF (see figure 8). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

3. Abbreviations used are defined in 6.1.

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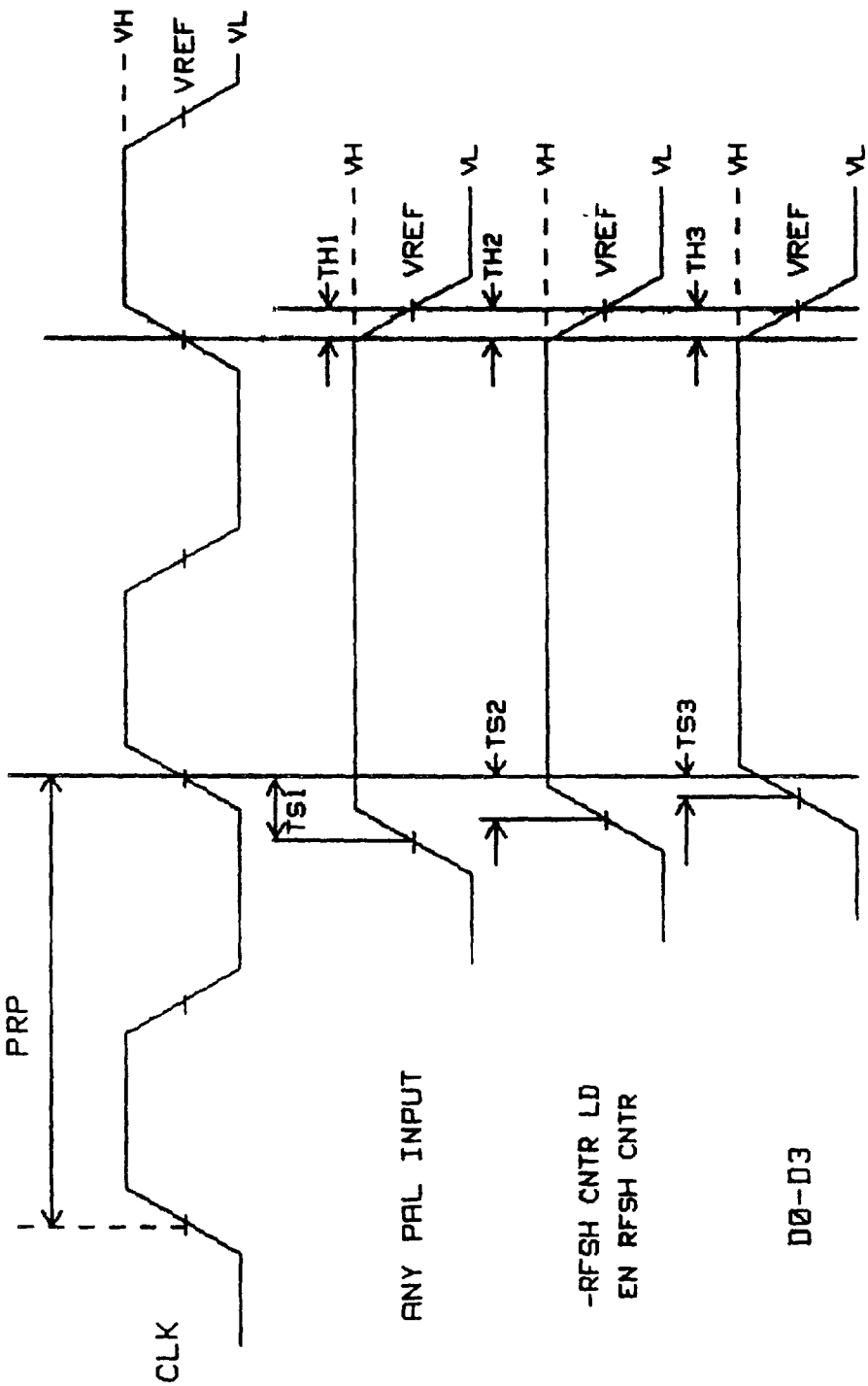


FIGURE 9. Input timing parameters for table XIII.

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NOTES:

1. The input parameters for figure 9 are:

   PRP = 84 ± 5 nS.
   TS1 = 60 nS minimum.
   TS2 = 55 nS minimum.
   TS3 = 35 nS minimum.
   TH1 = 30 nS minimum.
   TH2 = 40 nS minimum.
   TH3 = 30 nS minimum.
   VREF = 1.3 ± 0.1 V.

2. Transition times shall be 15 ± 1.5 nS.

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TABLE XIV. Dynamic test.

Test cond	Test description	Notes
N1	Dynamic test, sample maximum pattern rate minimum power supply	1-7
N2	Dynamic test, sample maximum pattern rate maximum power supply	1-7

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	N1 N2
Power supply current limit	0.868 A	NA	N1,N2
Input voltage high, (VH) low, (VL)	 2.4 V 0.5 V	 ± 120 mV ± 25 mV	 N1,N2 N1,N2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 N1 N2 N1,N2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. The load circuit for this test is specified on figure 8.

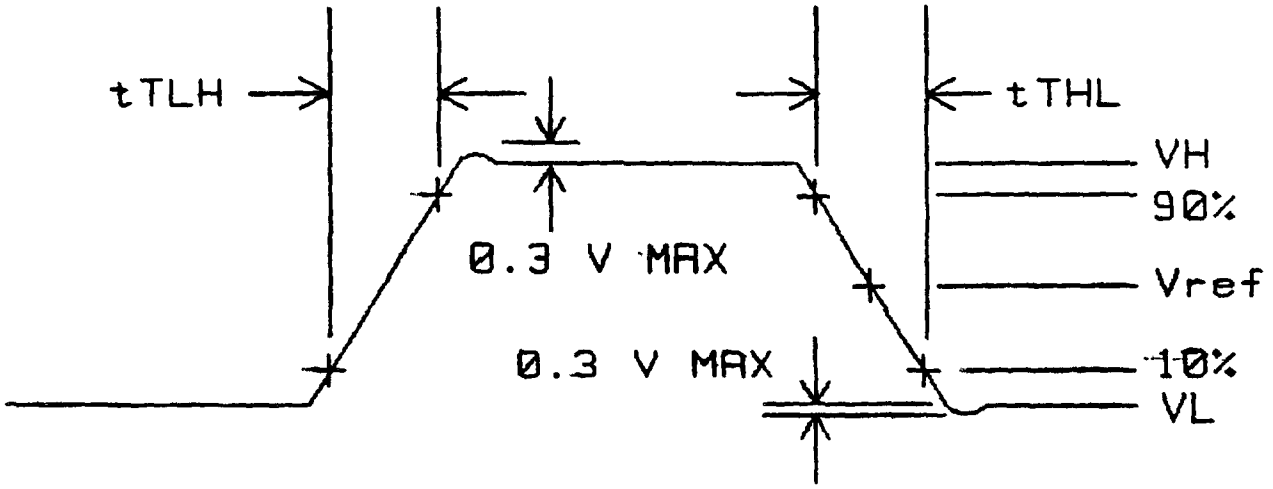
4. All of the input waveforms shall meet the requirements shown on figure 10.

5. This test shall be performed as indicated in table XIV. The dynamic parameters are found in table XV.

6. Input/Output (I/O) waveforms are as referenced on figure 11.

7. Abbreviations used are defined in 6.1.

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NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.5 V.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
22,21,70,74	2.5 ± 1.5 nS	2.5 ± 1.5 nS
96,83,53,97,52,98,47,24,77, 17,3,66,13,68,16,38,37,87,36, 46,32,41,91	15.0 ± 1.5 nS	6.0 ± 1.5 nS

FIGURE 10. Sample input waveform for table XIV.

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TABLE XV. Dynamic parameters.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
25	1	CLK to TCLK, tPLH	-	-	-	38	38	38	nS	11c
26	2	CLK to TCLK, tPHL	-	-	-	38	38	38	nS	11c
1	3	CLK cycle time	84	84	84	-	-	-	nS	11a, 11b, 11c, 11d
2	4	CLK high width	42	42	42	-	-	-	nS	11a, 11b, 11c, 11d
3	5	CLK low width	42	42	42	-	-	-	nS	11a, 11b, 11c 11d
4	6	-BES setup time 11/	30	30	30	-	-	-	nS	11a, 11b
5	7	CLK (RE) to -EXT REQ DET, tPHL 1/, 2/ (read or write)	-	-	-	58	58	56	nS	11a, 11b
6	8	CLK (RE) to -RAS low, tPHL (read or write) 1/, 2/	-	-	-	56	56	56	nS	11a, 11b
18	9	CLK (RE) to -DTACK low, tPHL (read) 1/, 2/, 9/	-	-	-	56	56	56	nS	11b
8	10	CLK (RE) to -DTACK low, tPHL (write) 1/, 2/, 10/	-	-	-	58	58	56	nS	11a
7	11	CLK (RE) to MUX SELECT, tPHL (read or write) 1/, 2/	-	-	-	56	56	56	nS	11a, 11b
9	13	CLK (RE) to -UB CAS low, tPHL (read or write) 1/, 2/, 3/	-	-	-	56	56	56	nS	11a, 11b

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TABLE XV. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
8	14	CLK (RE) to -LB CAS low, tPHL (read or write) 1/, 2/, 4/	-	-	-	56	56	56	nS	11a, 11b
9	15	CLK (RE) to -LATCH low, tPHL (read or write) 1/, 2/	-	-	-	56	56	56	nS	11a, 11b
10	16	CLK (RE) to MUX SELECT, tPHL (read or write) 1/, 2/	-	-	-	61	56	61	nS	11a, 11b
9	17	CLK (RE) to -RAS high, tPLH (read or write) 1/, 2/	-	-	-	61	56	61	nS	11a, 11b
11	18	CLK (RE) to -UB CAS high, tPLH (read or write) 1/, 2/, 3/	-	-	-	61	56	61	nS	11a, 11b
11	19	CLK (RE) to -LB CAS, tPLH (read or write) 1/, 2/, 4/	-	-	-	61	56	61	nS	11a, 11b
19	20	CLK (RE) to -IN LATCH, tPLH (read) 1/, 2/	-	-	-	86	86	86	nS	11b
12	21	CLK (RE) to -IN LATCH, tPLH (write) 1/, 2/	-	-	-	86	86	86	nS	11a
20	22	CLK (RE) to -DATA EN OUT, tPHL (read) 1/, 2/	-	-	-	56	56	56	nS	11b

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TABLE XV. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
13	23	CLK (RE) to -EXT REQ DET, tPLH (read or write) 1/, 2/	-	-	-	61	56	61	nS	11a, 11b
13	24	CLK (RE) to -DTACK high, tPLH (read or write) 1/, 2/	-	-	-	61	56	61	nS	11a, 11b
13	25	CLK (RE) to -LATCH high, tPLH (read or write) 1/, 2/	-	-	-	61	56	61	nS	11a, 11b
21	26	CLK (RE) to -IN LATCH low, tPHL (read or write) 1/, 2/	-	-	-	92	87	92	nS	11a, 11b
14	27	CLK (RE) to -LOCK high, tPLH (read or write) 1/, 2/	-	-	-	61	56	61	nS	11a, 11b
13	28	CLK (RE) to -DATA EN OUT, tPLH (read) 1/, 2/	-	-	-	61	56	61	nS	11b
15	29	CLK (RE) to -SINGLE ERR low, tPHL (read or write) 1/, 2/, 5/	-	-	-	56	56	56	nS	11a, 11b
15	30	CLK (RE) to -MULTI ERR low, tPHL (read or write) 1/, 2/, 6/	-	-	-	56	56	56	nS	11a, 11b
16	31	CLK (RE) to SINGLE ERR, tPLH (read or write) 1/, 2/, 6/	-	-	-	92	92	92	nS	11a, 11b

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TABLE XV. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
17	32	CLK (RE) to SINGLE ERR, tPHL (read or write) 1/, 2/, 6/	-	-	-	92	92	92	nS	11a, 11b
11	33	CLK (RE) -RFSH low, tPHL (read or write) 1/, 2/, 8/	-	-	-	56	56	56	nS	11a, 11b
22	34	CLK (RE) to -SINGLE ERR, tPLH (read or write) 1/, 2/, 6/	-	-	-	61	56	61	nS	11a, 11b
22	35	CLK (RE) to -MULTI ERR, tPLH (read or write) 1/, 2/, 7/	-	-	-	61	56	61	nS	11a, 11b
31	37	READ setup time, tS	30	30	30	-	-	-	nS	11d
31	38	-CS setup time, tS	30	30	30	-	-	-	nS	11d
31	39	-RD DTACK EN setup time, tS	30	30	30	-	-	-	nS	11d
31	40	-WT DTACK EN setup time, tS	30	30	30	-	-	-	nS	11d
31	41	-RESET setup time, tS	30	30	30	-	-	-	nS	11d
31	42	-LDS setup time, tS	30	30	30	-	-	-	nS	11d
31	43	-UDS setup time, tS	30	30	30	-	-	-	nS	11d
31	44	-LB SNGLE ERR setup time, tS	30	30	30	-	-	-	nS	11d

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TABLE XV. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
31	45	-UB SNGLE ERR setup time, tS	30	30	30	-	-	-	nS	11d
31	46	-LB MULTI ERR setup time, tS	30	30	30	-	-	-	nS	11d
31	47	-UB MULTI ERR setup time, tS	30	30	30	-	-	-	nS	11d
32	48	-RFSH CNTR LD setup time, tS	25	25	25	-	-	-	nS	11d
32	49	EN RFSH CNTR setup time, tS	25	25	25	-	-	-	nS	11d
33	50-53	D0-D3 setup time, tS	7	7	7	-	-	-	nS	11d
34	54-61	RFSH CNT 0-7 setup time, tS	20	20	20	-	-	-	nS	11d
35	62	READ hold time, tH	36	36	36	-	-	-	nS	11d
35	63	-CS hold time, tH	36	36	36	-	-	-	nS	11d
35	64	-RD DTACK EN hold time, tH	36	36	36	-	-	-	nS	11d
35	65	-WT DTACK EN hold time, tH	36	36	36	-	-	-	nS	11d
35	66	-RESET hold time, tH	36	36	36	-	-	-	nS	11d
35	67	-LDS hold time, tH	36	36	36	-	-	-	nS	11d
35	68	-UDS hold time, tH	36	36	36	-	-	-	nS	11d

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TABLE XV. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Unit	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
35	69	-LB SNGLE ERR hold time, tH	36	36	36	-	-	-	nS	11d
35	70	-UB SNGLE ERR hold time, tH	36	36	36	-	-	-	nS	11d
35	71	-LB MULTI ERR hold time, tH	36	36	36	-	-	-	nS	11d
35	72	-UB MULTI ERR hold time, tH	36	36	36	-	-	-	nS	11d
36	73	-RFSH CNTR LD hold time, tH	46	46	46	-	-	-	nS	11d
36	74	EN RFSH CNTR hold time, tH	46	46	46	-	-	-	nS	11d
37	75-78	D0-D3 hold time, tH	41	41	41	-	-	-	nS	11d
38	79-86	RFSH CNT 0-7 hold time, tH	46	46	46	-	-	-	nS	11d
39	87	-BES hold time, tH	36	36	36	-	-	-	nS	11a, 11b
28	88-91	CLK (RE) to Q0-Q3 low	-	-	-	57	53	57	nS	11c
29	92-95	CLK (RE) to Q0-Q3 high	-	-	-	53	49	53	nS	11c
30, 40	96,97	CLK (RE) to TRFSH CNT TC1, TRFSH CNT TC2, tPLH	-	-	-	80	80	80	nS	11c
41	98	CLK (RE) to TRFSH DET tPLH	-	-	-	53	50	53	nS	11c

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NOTES:

1. (RE) = Rising Edge.

2. This parameter assumes that -CS is low during the cycle and meets the setup time with parameter 37 and the hold time of parameter 63.

3. This parameter assumes that -BES is asserted and meets setup conditions given by parameter 6.

4. This parameter assumes that -UDS is low during the cycle and meets the setup time with parameter 43 and the hold time of parameter 68.

5. This parameter assumes that -LDS is low during the cycle and meets the setup time with parameter 42 and the hold time with parameter 67.

6. This parameter assumes that either -LB SNGLE ERR or -UB SNGLE ERR is low. -LB SNGLE ERR must meet the setup time with parameter 44 and the hold time with parameter 69. -UB SNGLE ERR must meet the setup time with parameter 45 and the hold time with parameter 70.

7. This parameter assumes that either -LB MULTI ERR or -UB MULTI ERR is low. -LB MULTI ERR must meet the setup time with parameter 46 and the hold time with parameter 71. -UB MULTI ERR must meet the setup time with parameter 47 and the hold time with parameter 72.

8. This parameter assumes that a refresh cycle is in progress.

9. This parameter assumes that -RD DTACK EN is low during the cycle and meets the setup time with parameter 39 and the hold time with parameter 64.

10. This parameter assumes that -WT DTACK EN is low during the cycle and meets the setup time with parameter 40 and the hold time with parameter 65.

11. Timing relationships on figures 11a and 11b assume that -BES setup time is not so great as to have it recognized on a previous clock cycle.

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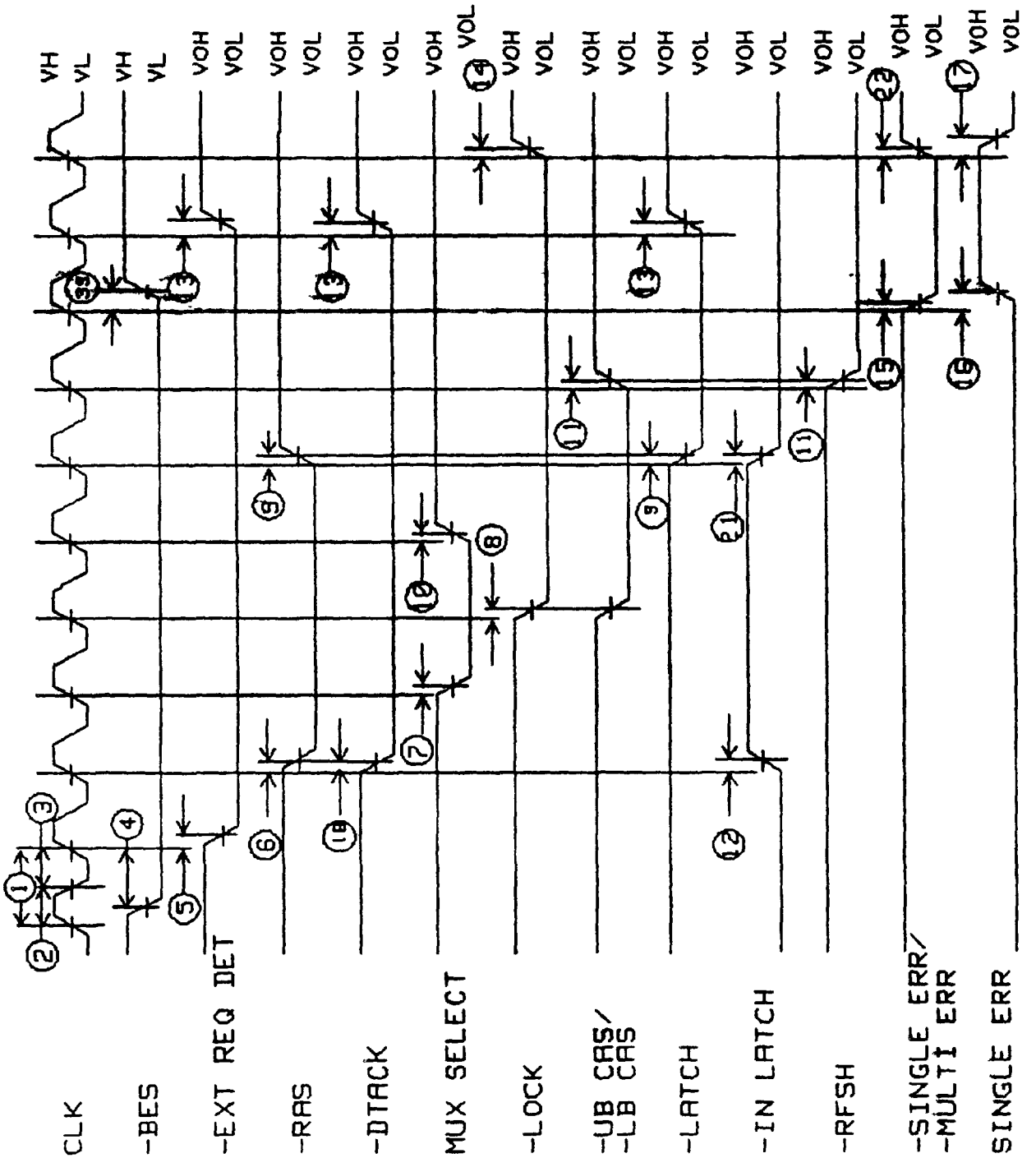


FIGURE 11a. Dynamic waveforms for table XV.

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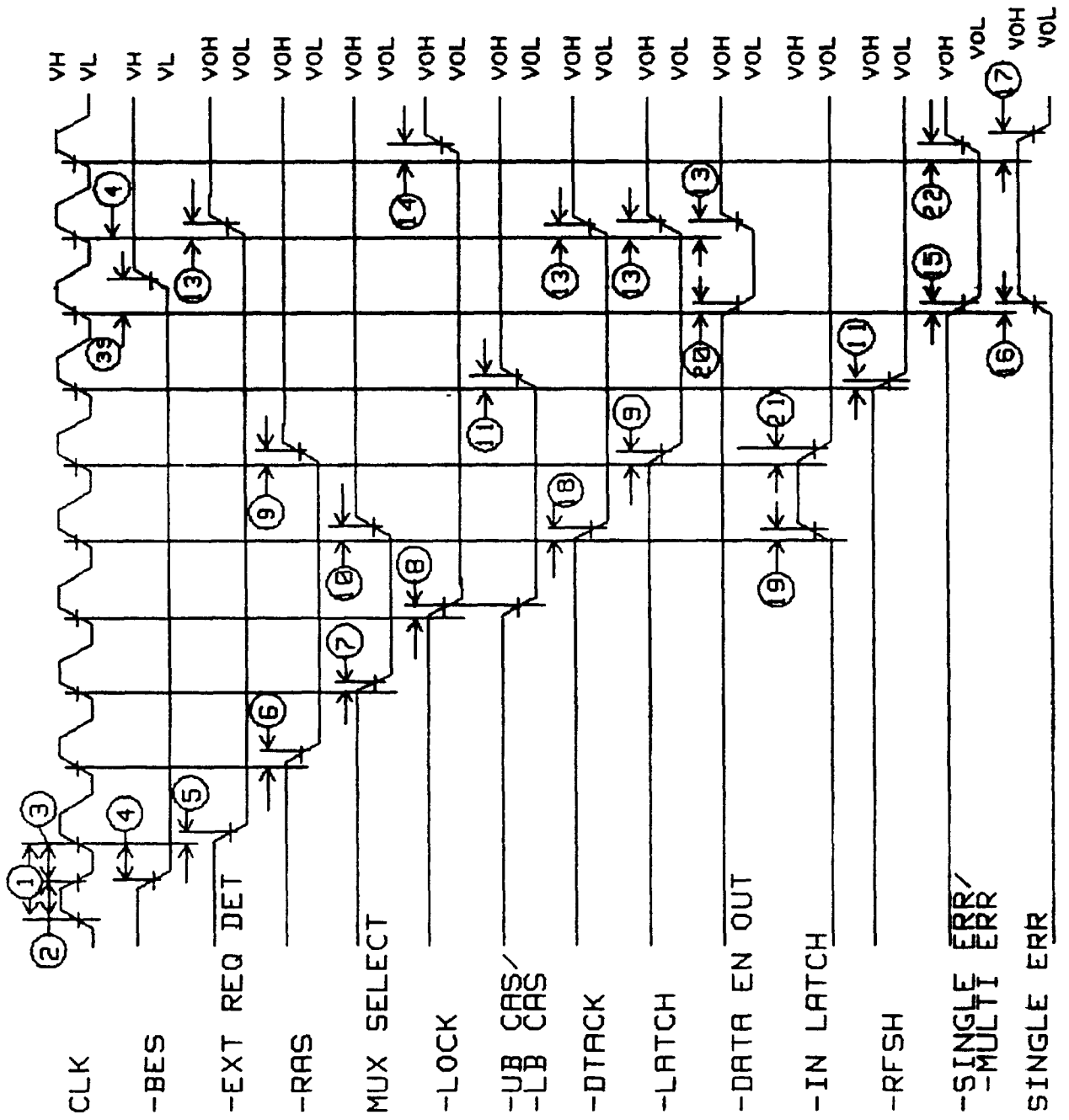


FIGURE 11b. Dynamic waveforms for table XV.

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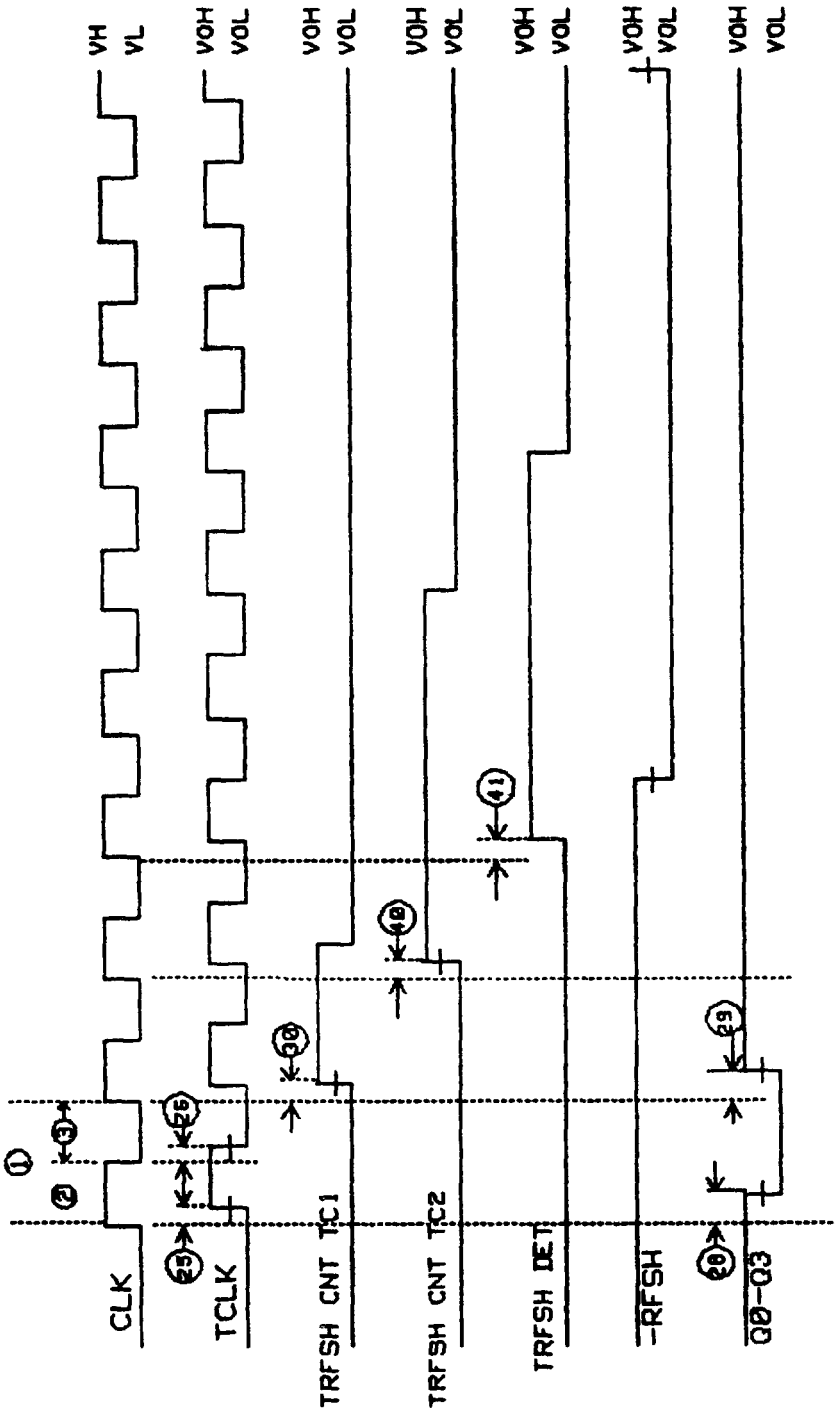


FIGURE 11c. Dynamic waveforms for table XV.

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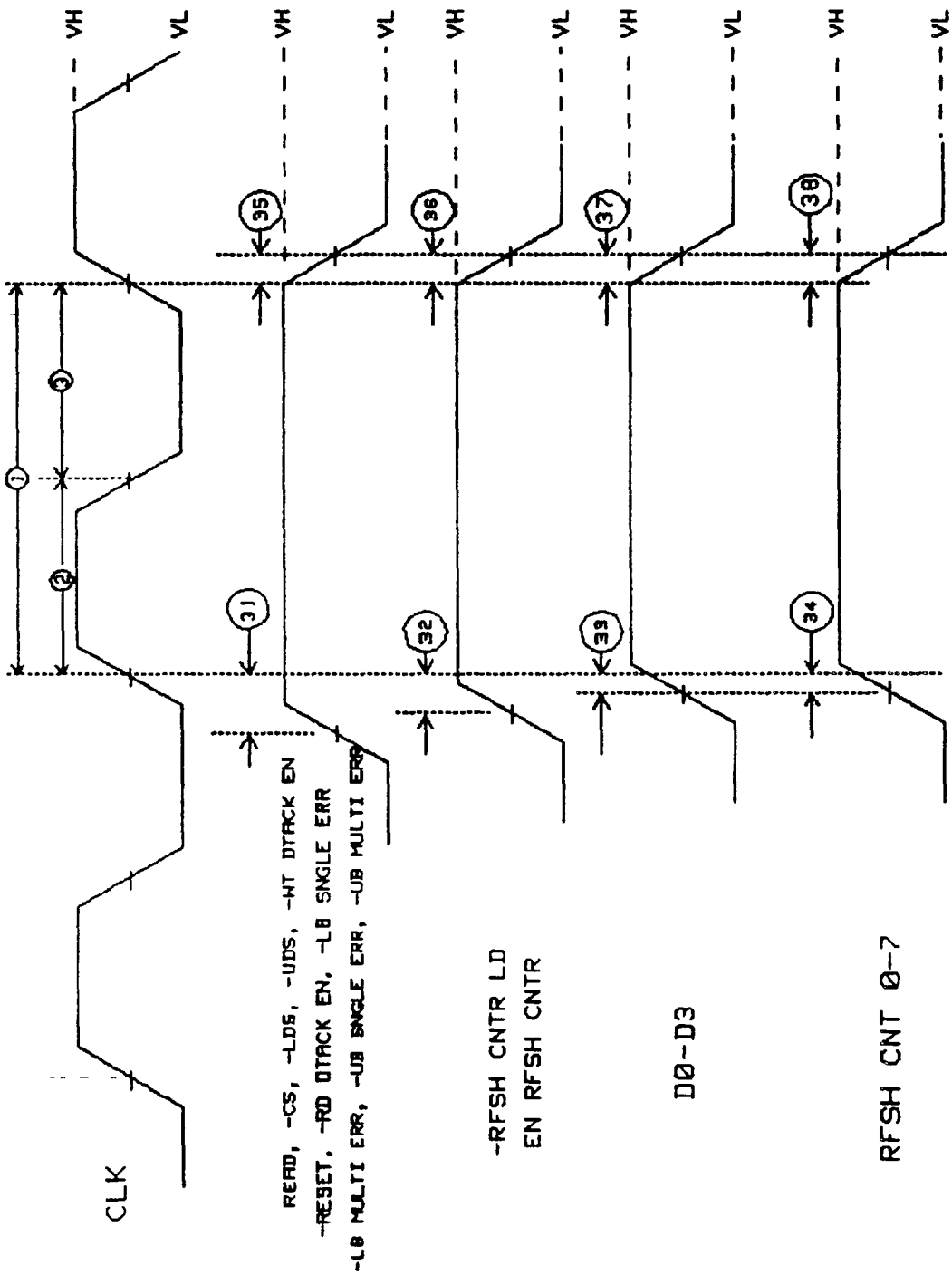


FIGURE 11d. Dynamic waveforms for table XV.

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TABLE XVI. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
P	PSFC	1,31, 51,81	3.9	6.5	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	P
Capacitance bridge voltage	0.3 Vp	± 30 mVp	P
Capacitance measurement	NA	± 130 nF	P

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVI.

3. Abbreviations used are defined in 6.1.

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TABLE XVII. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Q	PSTA	1	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	Q
Power supply current limit	0.868 A	NA	Q
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	Q Q
Power supply transient amplitude measurement	NA	± 20 mV	Q

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XVII.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XVII.

(d) Using the oscilloscope, D07, inspect for the most negative spike as shown on figure 12 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 12 pF. There shall be no other capacitance present on Vcc to ground.

3. The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

4. Abbreviations used are defined in 6.1.

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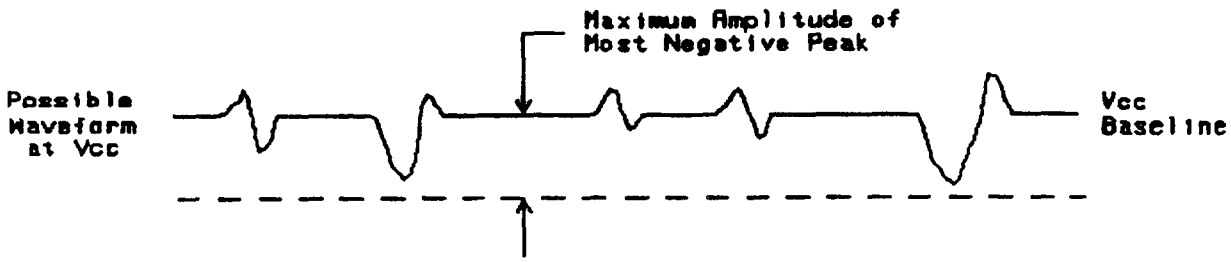


FIGURE 12. Power supply transient amplitude waveform for table XVII.

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TABLE XVIII. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs.
R1	Q0	18	0	500	mV	1-5	13,14
R2	Q1	71					
R3	Q2	20					
R4	Q3	73					
R5	-RFSH	99					
R6	-LOCK	49					
R7	-EXT REQ DET	92					
R8	-RAS	100					
R9	-DTACK	56					
R10	-IN LATCH	93					
R11	-LB CAS	6					
R12	-UB CAS	54					
R13	-LATCH	58					
R14	-DATA EN OUT	69					
R15	-SINGLE ERR	67					
R16	SINGLE ERR	72					
R17	-MULTI ERR	19					
R18	MUX SELECT	57	0	500	mV	1-5	13,14

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NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	R1-R18
Power supply current limit	0.868 A	NA	R1-R18
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 R1-R18 R1-R18
Output crosstalk measurement	NA	± 25 mV	R1-R18

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

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NOTES: - Continued.

(d) For each output under test in table XVIII, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 13.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 8.

4. The test circuit shall be as indicated on figure 14. Module inputs shall not violate the minimum input timing parameters referenced on figure 9.

5. Abbreviations used are defined in 6.1.

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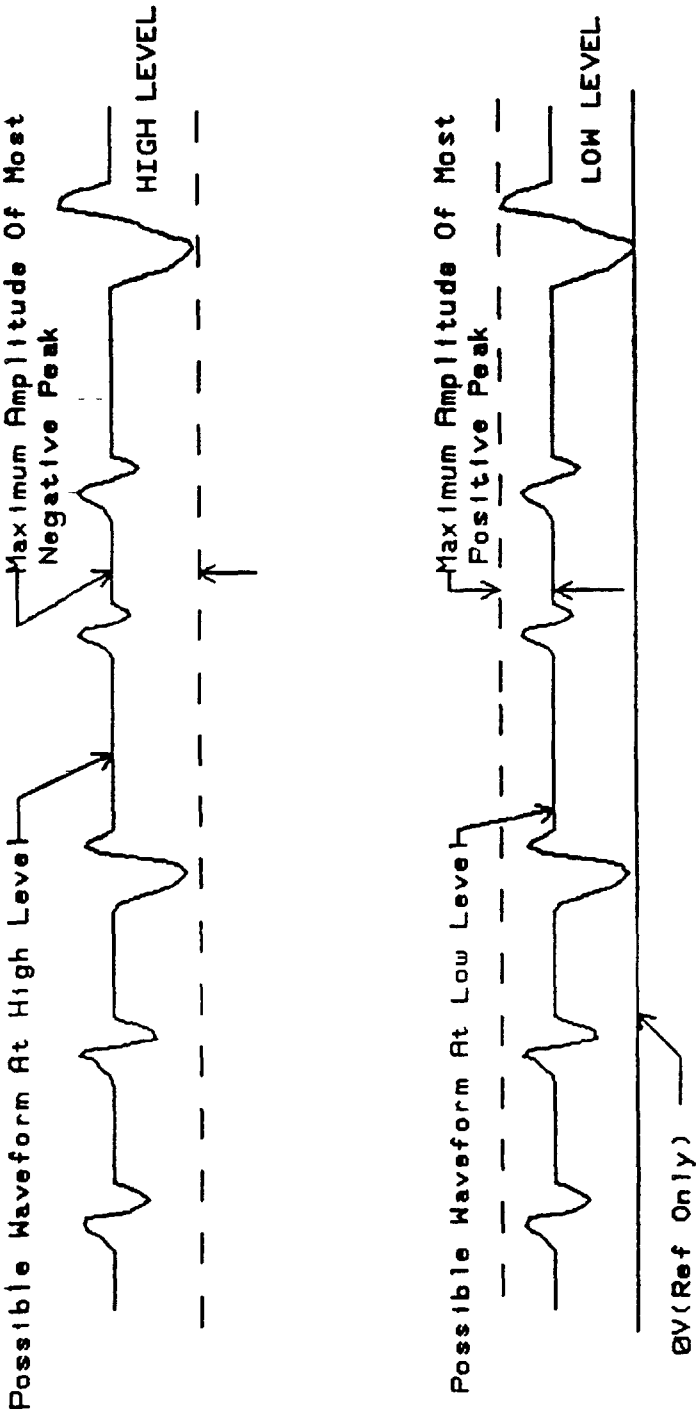


FIGURE 13. Output crosstalk, high and low level, test waveforms for tables XVIII and XIX.

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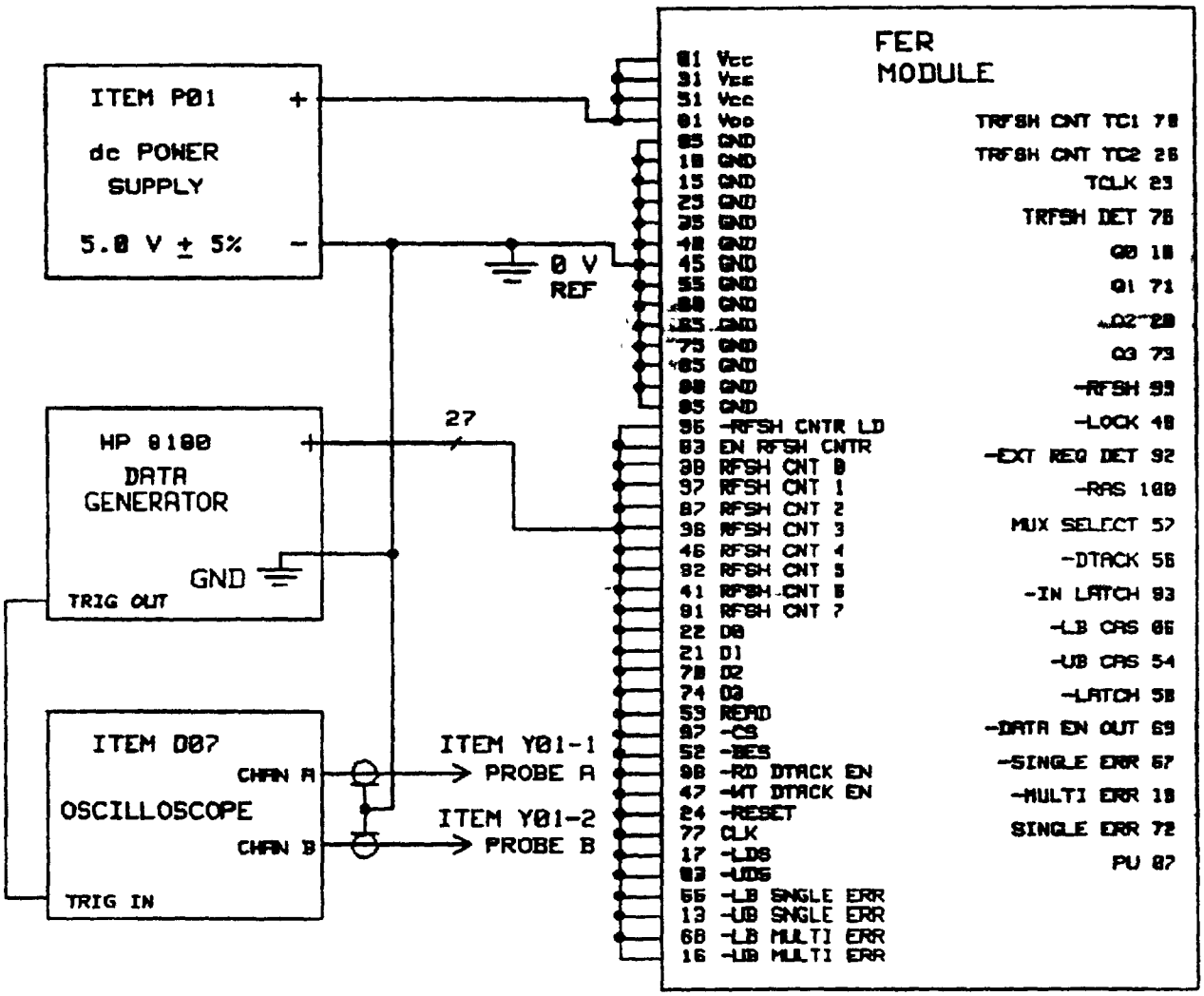


FIGURE 14. Output crosstalk, high and low level, test circuit for tables XVIII and XIX.

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TABLE XIX. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs.
S1	Q0	18	0	750	mV	1-5	13,14
S2	Q1	71					
S3	Q2	20					
S4	Q3	73	0	750	mV	1-5	13,14
S5	-RFSH	99	0	500	mV	1-5	13,14
S6	-LOCK	49					
S7	-EXT REQ DET	92					
S8	-RAS	100					
S9	-DTACK	56					
S10	-IN LATCH	93					
S11	-LB CAS	6					
S12	-UB CAS	54					
S13	-LATCH	58					
S14	-DATA EN OUT	69					
S15	-SINGLE ERR	67					
S16	SINGLE ERR	72					
S17	-MULTI ERR	19					
S18	MUX SELECT	57	0	500	mV	1-5	13,14

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NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	S1-S18
Power supply current limit	0.868 A	NA	S1-S18
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S1-S18 S1-S18
Output crosstalk measurement	 NA NA	 ± 37.5 mV ± 25.0 mV	 S1-S4 S5-S18

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

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NOTES: - Continued.

(d) For each output under test in table XIX, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 13.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 8.

4. The test circuit shall be as indicated on figure 14. The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

5. Abbreviations used are defined in 6.1.

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TABLE XX. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	-UDS	3	2	25	pF	1-3
T2	RFSH CNT 3	36				
T3	RFSH CNT 1	37				
T4	RFSH CNT 0	38				
T5	RFSH CNT 2	87				
T6	EN RFSH CNTR	83				
T7	-WT DTACK EN	47				
T8	-CS	97	2	25	pF	1-3
T9	RFSH CNT 4	46	2	30	pF	1-3
T10	RFSH CNT 5	32				
T11	RFSH CNT 6	41				
T12	RFSH CNT 7	91	2	30	pF	1-3
T13	D0	22	2	20	pF	1-3
T14	D1	21				
T15	D2	70				
T16	D3	74				
T17	-UB SNGLE ERR	13				
T18	-LDS	17				
T19	-LB SNGLE ERR	66	2	20	pF	1-3
T20	-LB MULTI ERR	68	2	31	pF	1-3

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TABLE XX. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T21	CLK	77	2	31	pF	1-3
T22	-BES	52	4	80	pF	1-3
T23	READ	53	4	80	pF	1-3
T24	-RESET	24	2	35	pF	1-3
T25	-RD DTACK EN	98				
T26	-UB MULTI ERR	16	2	35	pF	1-3
T27	-RFSH CNTR LD	96	4	60	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	T1-T27
Capacitance meter voltage	15 mV	± 1.5 mV	T1-T27
Capacitance measurement	NA	± 1.15 pF	T1-T8
	NA	± 1.4 pF	T9-T12
	NA	± 0.9 pF	T13-T19
	NA	± 1.45 pF	T20,T21
	NA	± 3.8 pF	T22,T23
	NA	± 1.65 pF	T24-T26
	NA	± 2.8 pF	T27

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XX.

3. Abbreviations used are defined in 6.1.

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TABLE XXI. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
U1	TCLK	23	4	140	pF	1-3
U2	TRFSH DET	76	2	45	pF	1-3
U3	TRFSH CNT TC1	78				
U4	SINGLE ERR	72	2	45	pF	1-3
U5	-LB CAS	6	2	30	pF	1-3
U6	-MULTI ERR	19				
U7	-LATCH	58				
U8	-DATA EN OUT	69	2	30	pF	1-3
U9	-LOCK	49	2	20	pF	1-3
U10	-RFSH	99				
U11	-RAS	100				
U12	Q0	18				
U13	Q1	71				
U14	Q2	20				
U15	Q3	73	2	20	pF	1-3
U16	TRFSH CNT TC2	26	2	38	pF	1-3
U17	-UB CAS	54	2	35	pF	1-3
U18	-IN LATCH	93	2	35	pF	1-3
U19	-SINGLE ERR	67	2	25	pF	1-3
U20	-EXT REQ DET	92	2	30	pF	1-3

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TABLE XXI. Output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
U21	-DTACK	56	2	60	pF	1-3
U22	MUX SELECT	57	2	70	pF	1-3

NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	U1-U22
Capacitance meter voltage	15 mV	± 1.5 mV	U1-U22
Capacitance measurement	NA	± 6.8 pF	U1
	NA	± 2.15 pF	U2-U4
	NA	± 1.4 pF	U5-U8
	NA	± 0.9 pF	U9-U15
	NA	± 1.8 pF	U16
	NA	± 1.65 pF	U17,U18
	NA	± 1.15 pF	U19
	NA	± 1.4 pF	U20
	NA	± 2.9 pF	U21
	NA	± 3.4 pF	U22

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXI.

3. Abbreviations used are defined in 6.1.

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TABLE XXII. Threshold test.

Test cond	Parameter	Notes	Fig
V	Threshold	1-4	9

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	V
Power supply current limit	0.868 A	NA	V
Pattern voltage high, (VH) low, (VL)	 See note 3 See note 3		 V V
Output voltage measurement high (Voh) low (Vol)	 NA NA	 ± 105 mV ± 25 mV	 V V

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results to the output low and high levels specified in the output voltage tests. The module inputs shall not violate the minimum input timing parameters referenced on figure 9.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. The input threshold levels appear in the output voltage tests.

4. Abbreviations used are defined in 6.1.

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TABLE XXIII. Output crosstalk duration test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
W1	Q0	18	0	59	nS	1-4	15
W2	Q1	71					
W3	Q2	20					
W4	Q3	73	0	59	nS	1-4	15

NOTES:

1. Output crosstalk duration test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 50 V	W1-W4
Power supply current limit	0.868 A	NA	W1-W4
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 W1-W4 W1-W4
Crosstalk duration measurement	NA	± 2.95 nS	W1-W4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

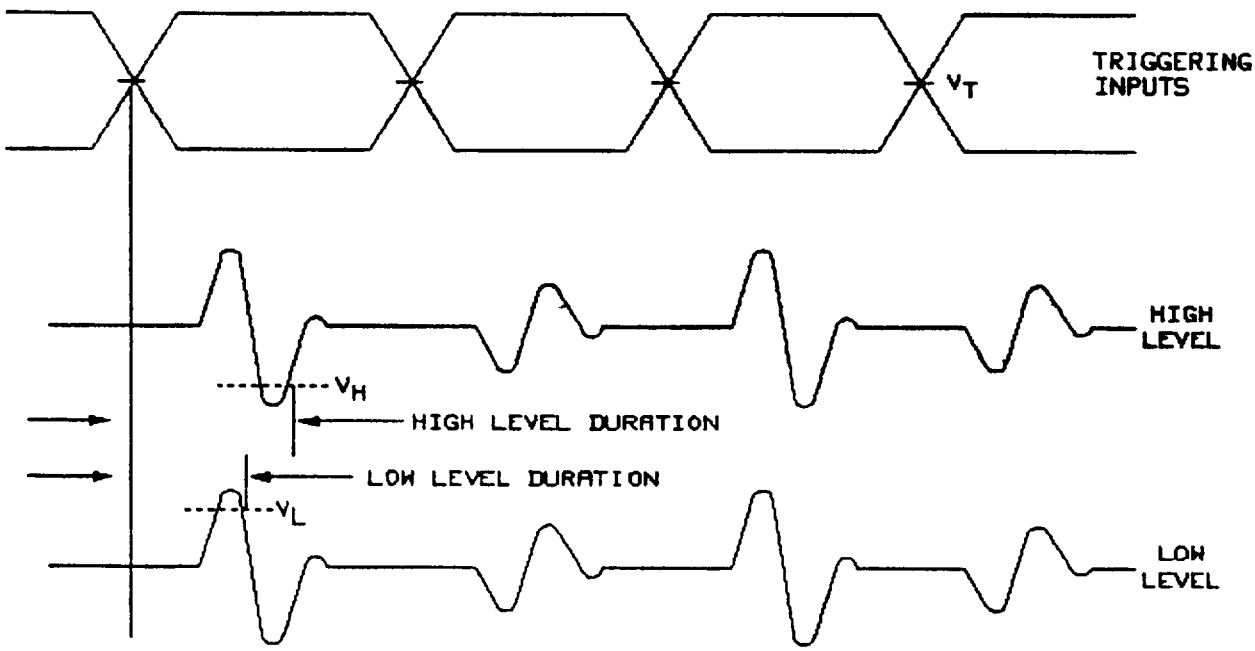
(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) The duration shall be measured from the triggering edge of the input as shown on figure 15.

3. All outputs shall be loaded as specified on figure 8.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. Vt = 1.5 V.

2. VH = VL = 500 mV.

3. The crosstalk duration is measured from the triggering edge of the input to the point where output crosstalk returns to the specified voltage. If the crosstalk amplitude is smaller than the duration measurement voltage (VH or VL), the crosstalk duration shall be recorded as 0 nS.

FIGURE 15. Output crosstalk duration test waveforms for table XXIII.

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